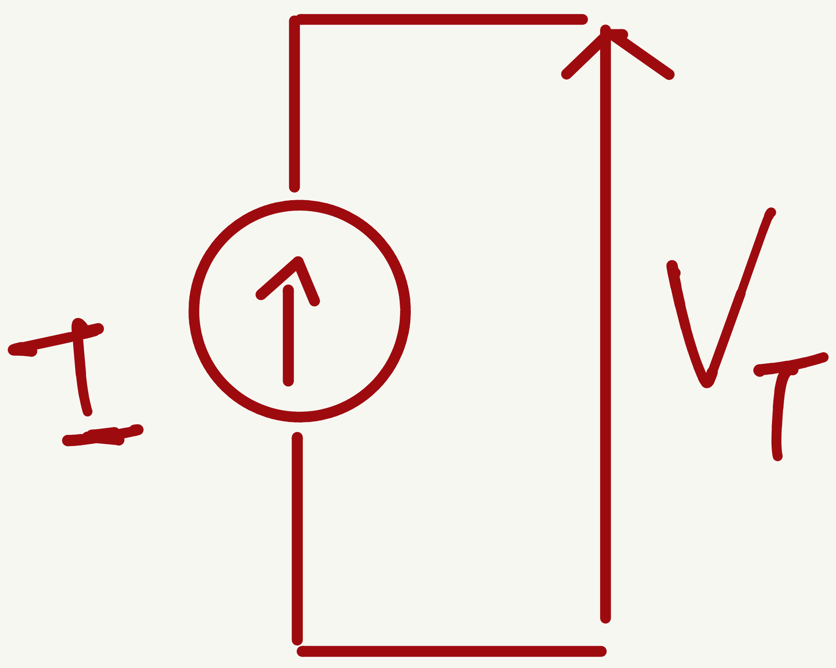
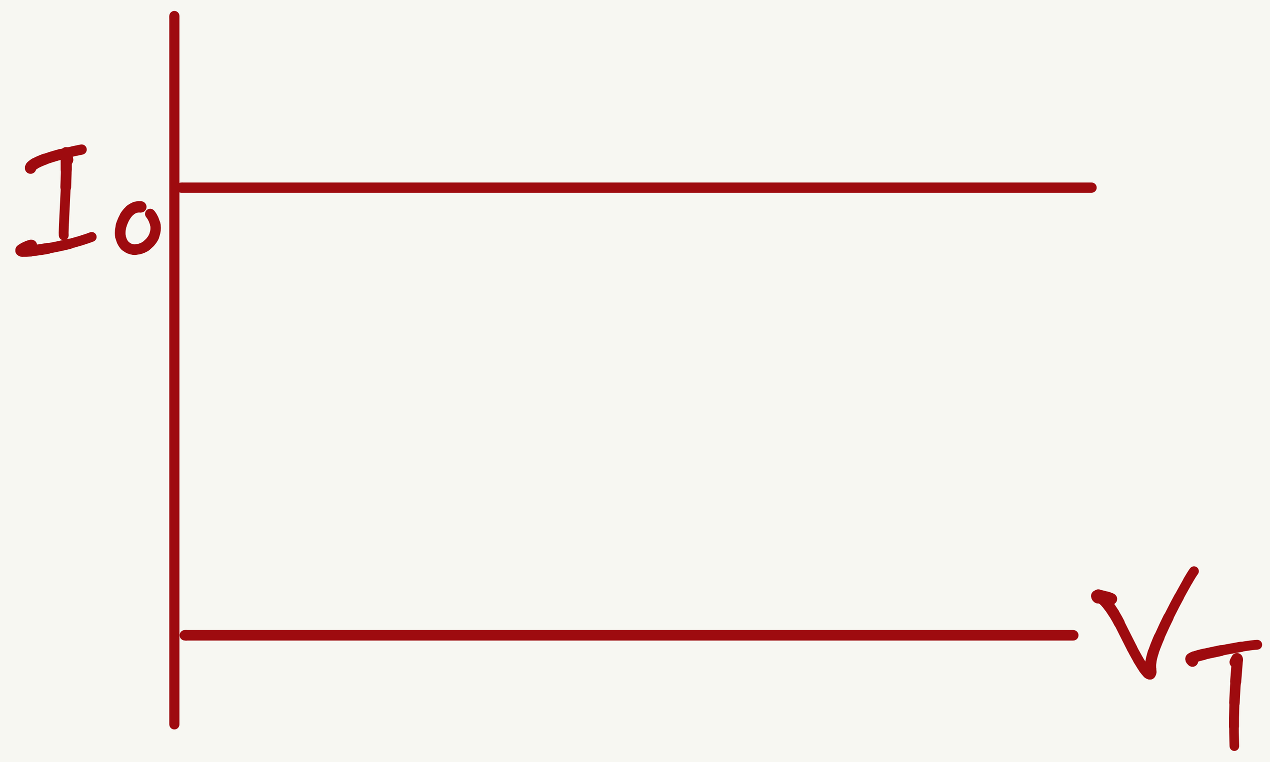
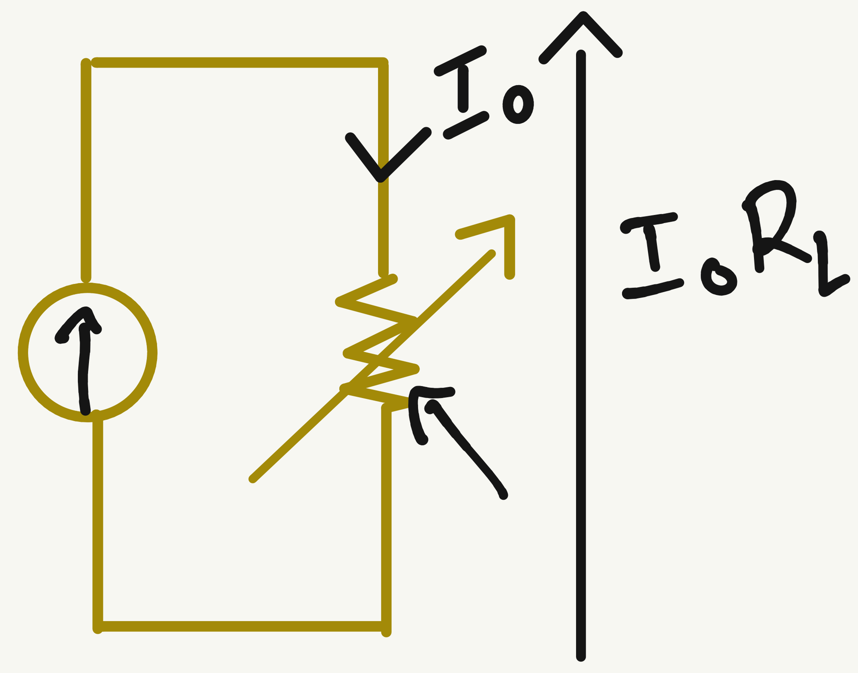
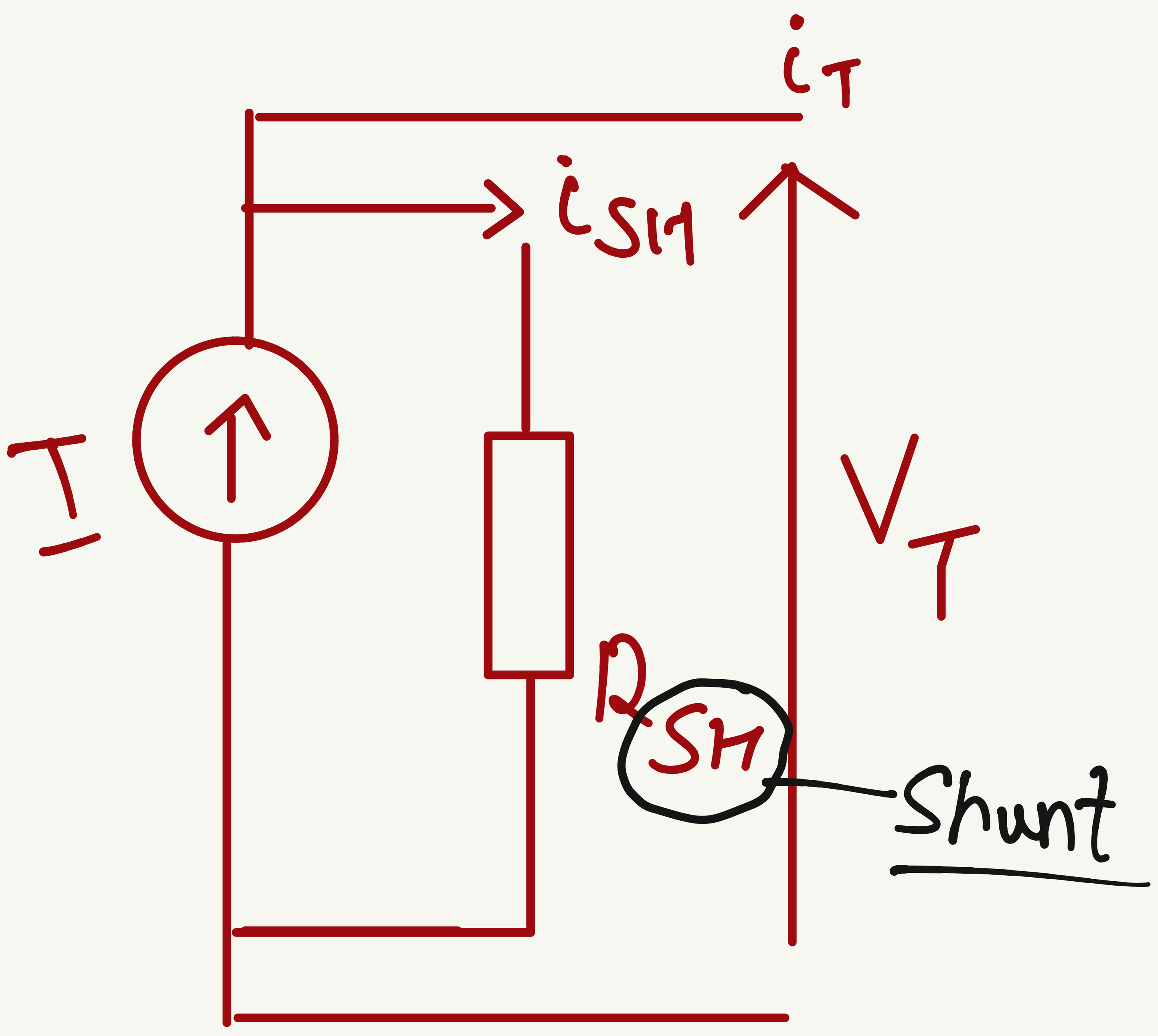
Ideal current source







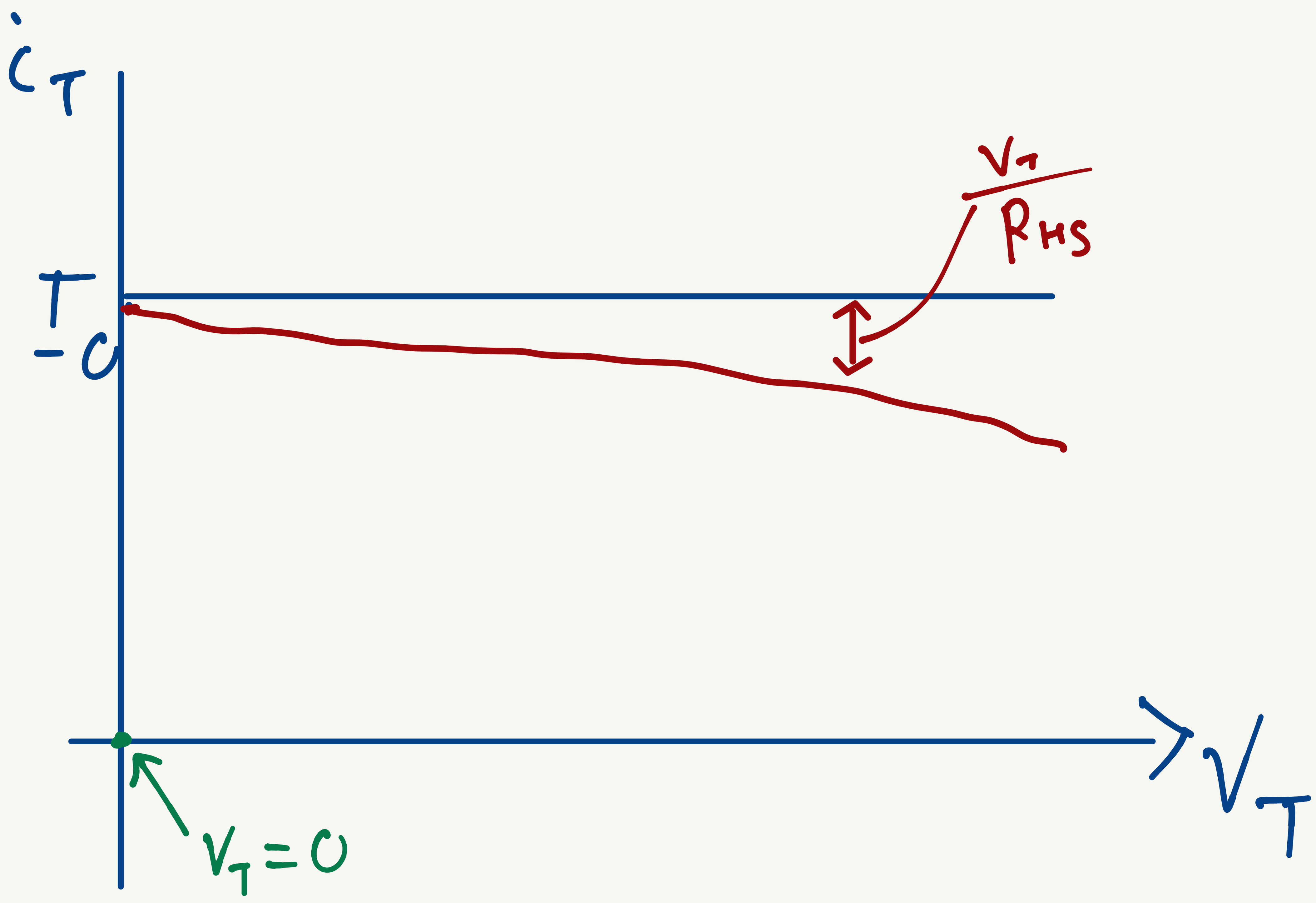
Practical current source



$$I = i_T + i_{SH}$$

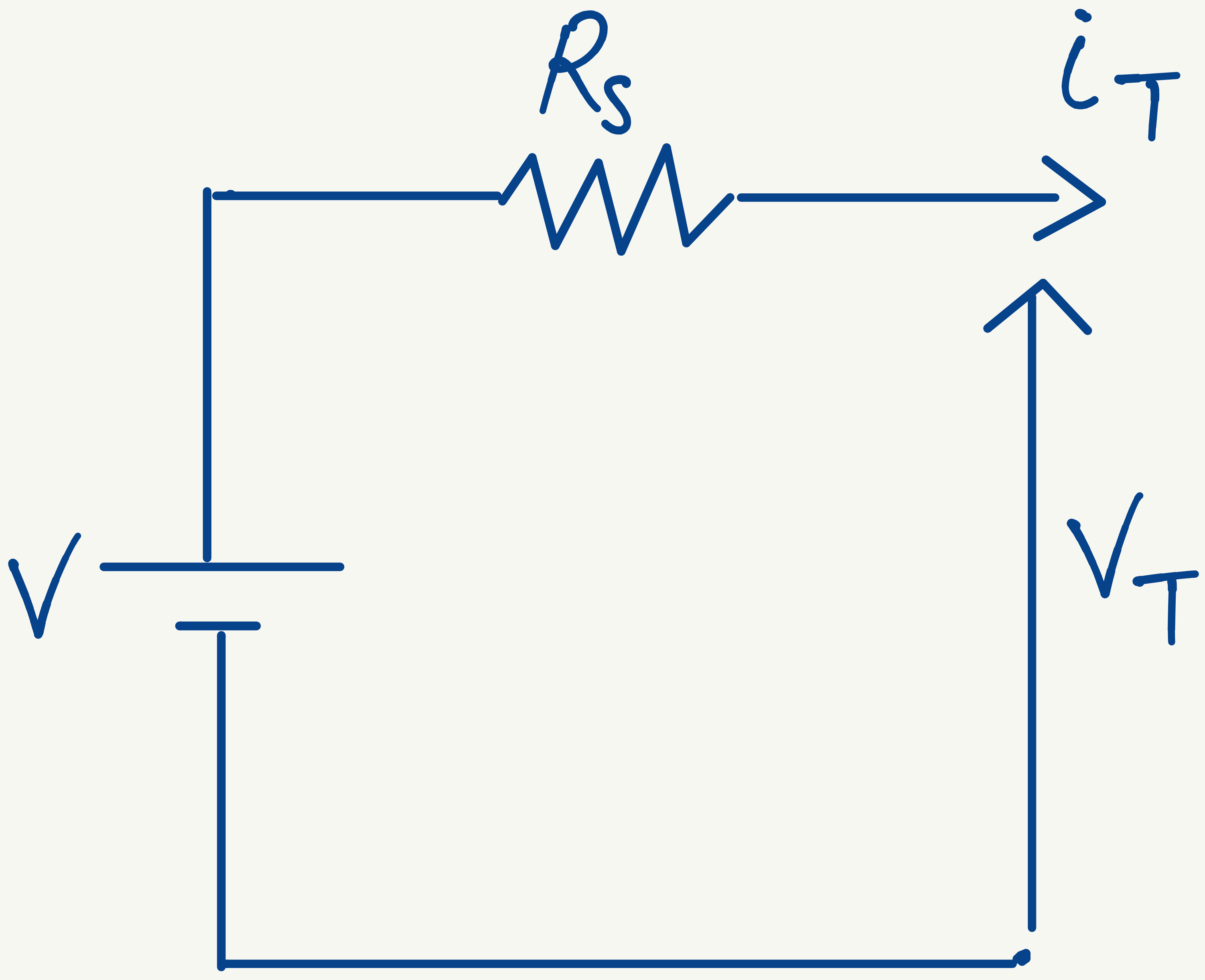
$$I = i_T + \frac{V_T}{R_{HS}}$$

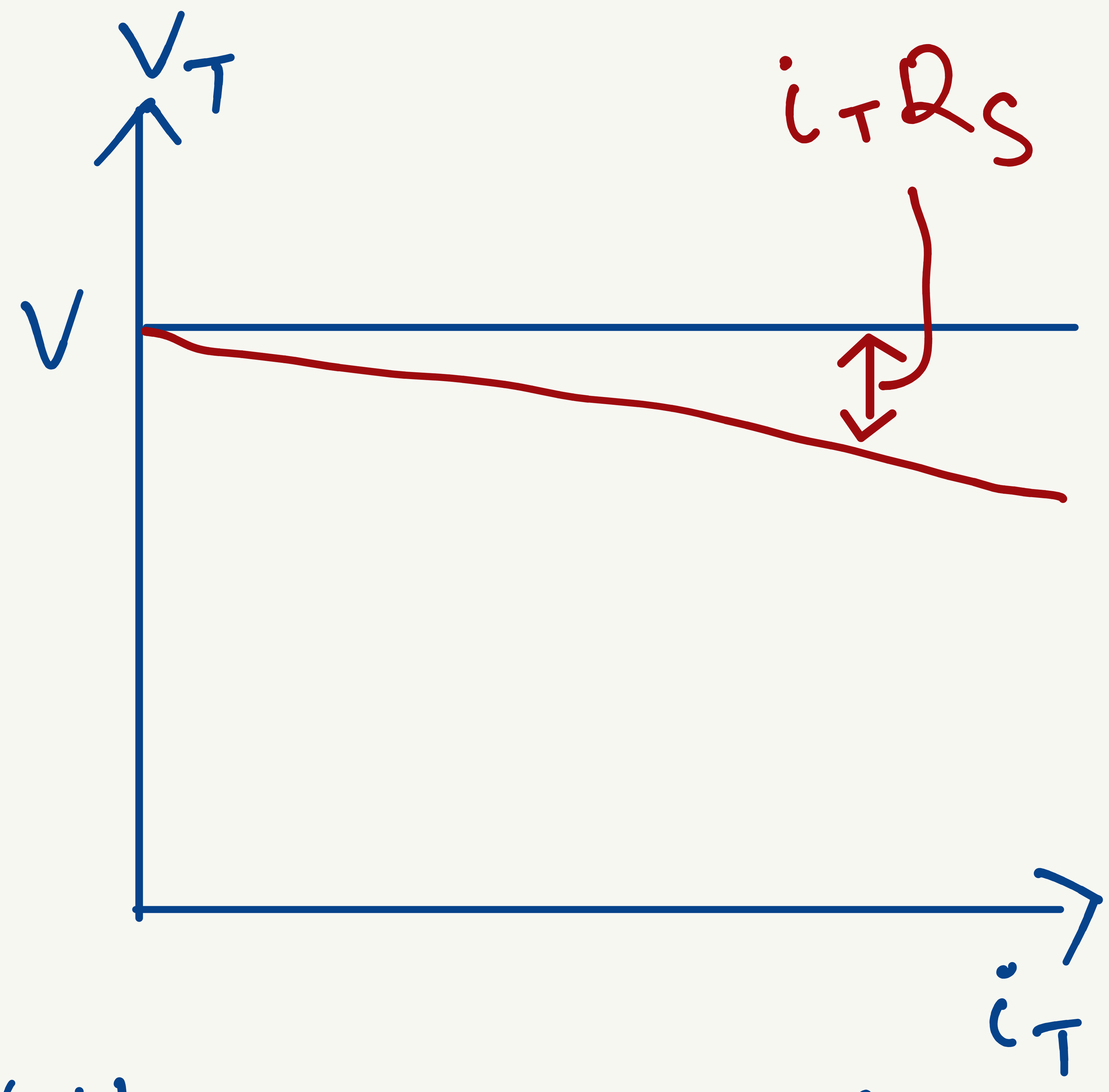
$$i_T = I - \boxed{\frac{V_T}{R_{HS}}}$$



① Example

Battery ⟶ Voltage source (practical)





* Power supply (AC) ⟶ Voltage source (practical)

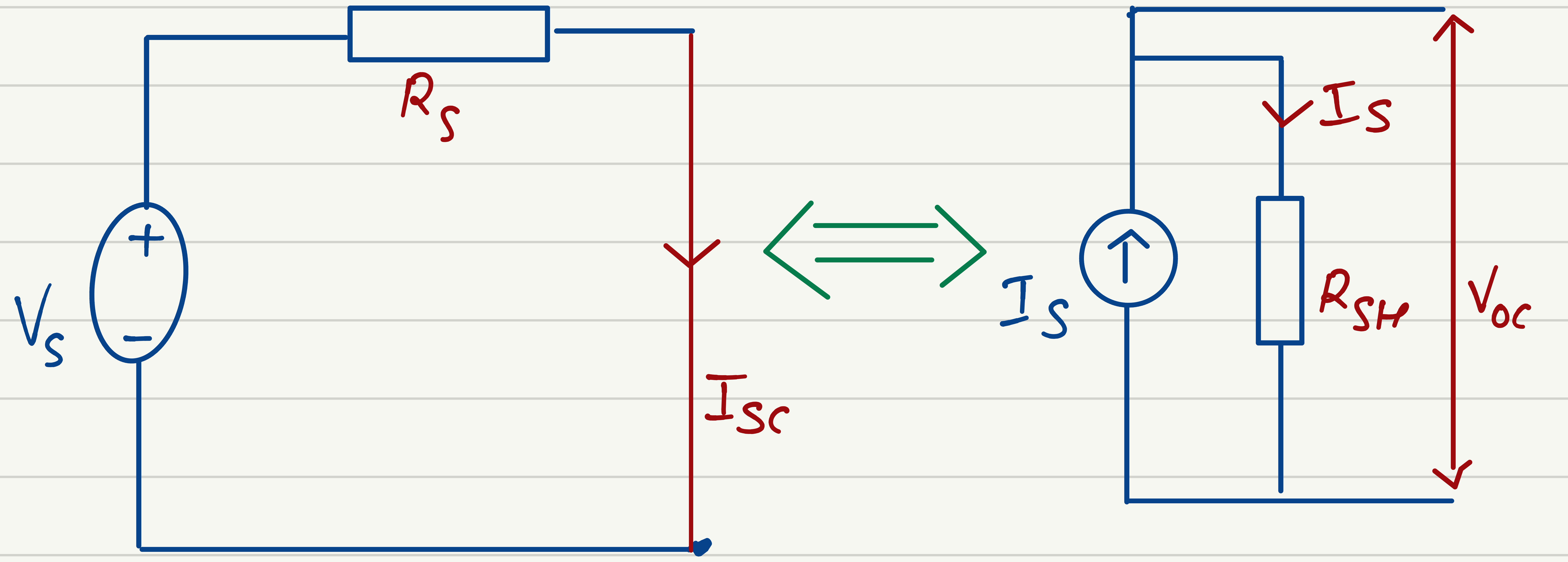
* Solar source ⟶ practical current source.

which on → voltage source
which on → current source

① No superconductor.

* Source Transformation

① Valid only for practical source.



$$I_{sc} = \frac{V_s}{R_s} = I_s$$

&

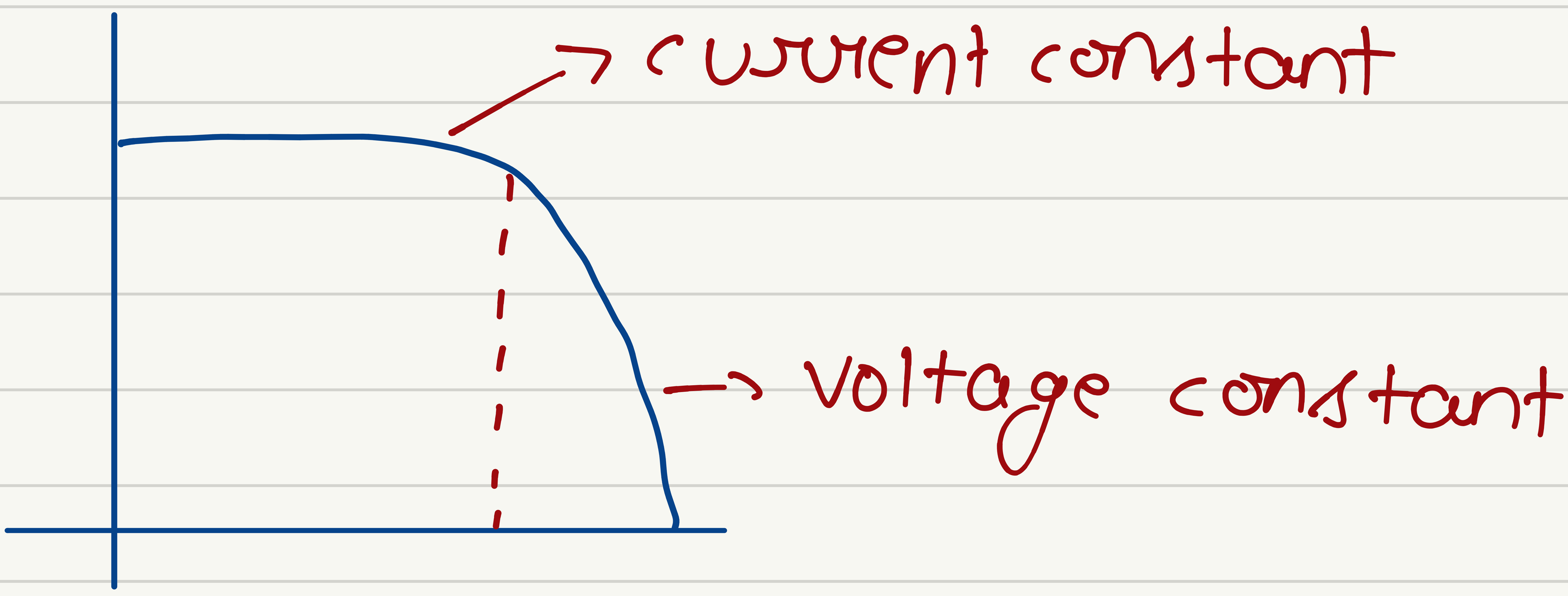
$$R_s = R_{SH} \; (\parallel)$$

parallel

$$V_{oc} = I_s R_{SH} = V_s$$

&

$$R_s = R_{SH} \; (\text{series})$$



* Dependent sources (controlled sources)

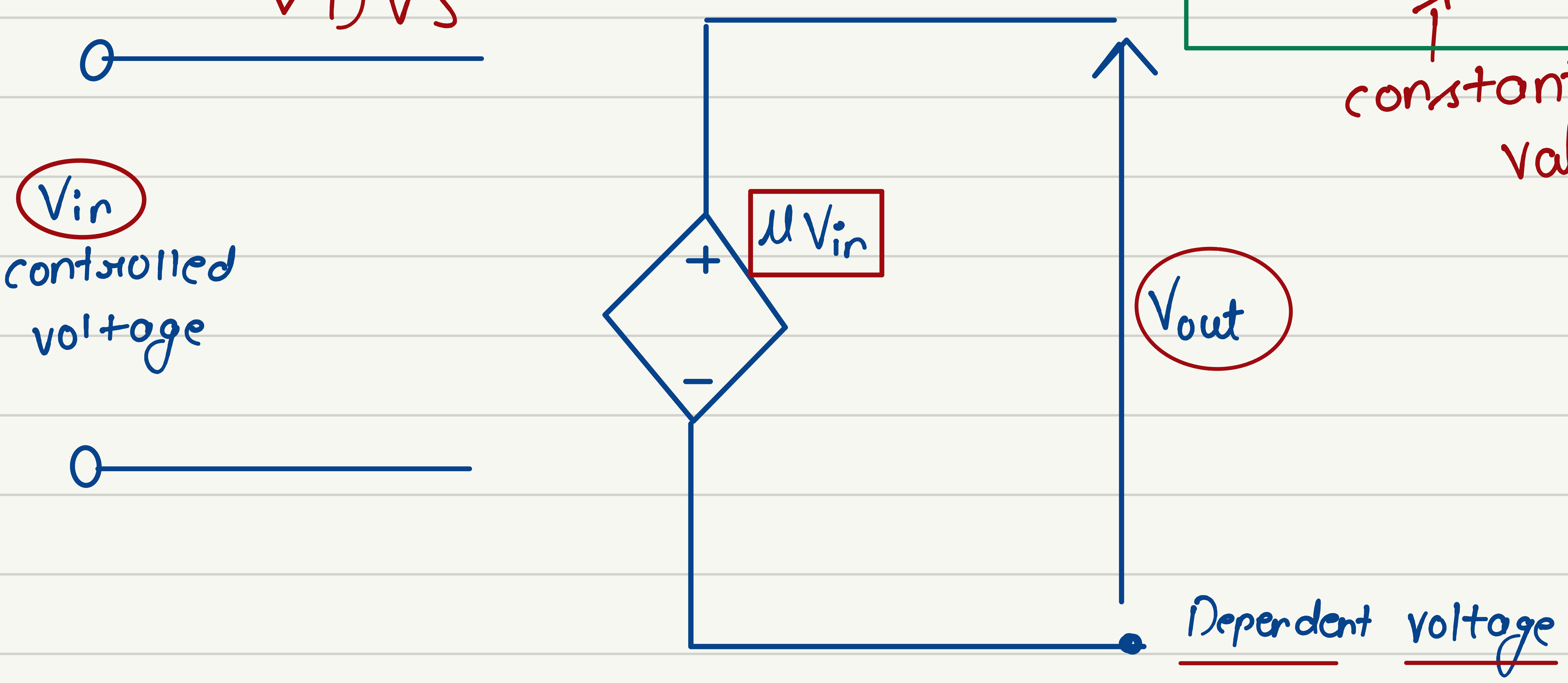
① Voltage dependent voltage source

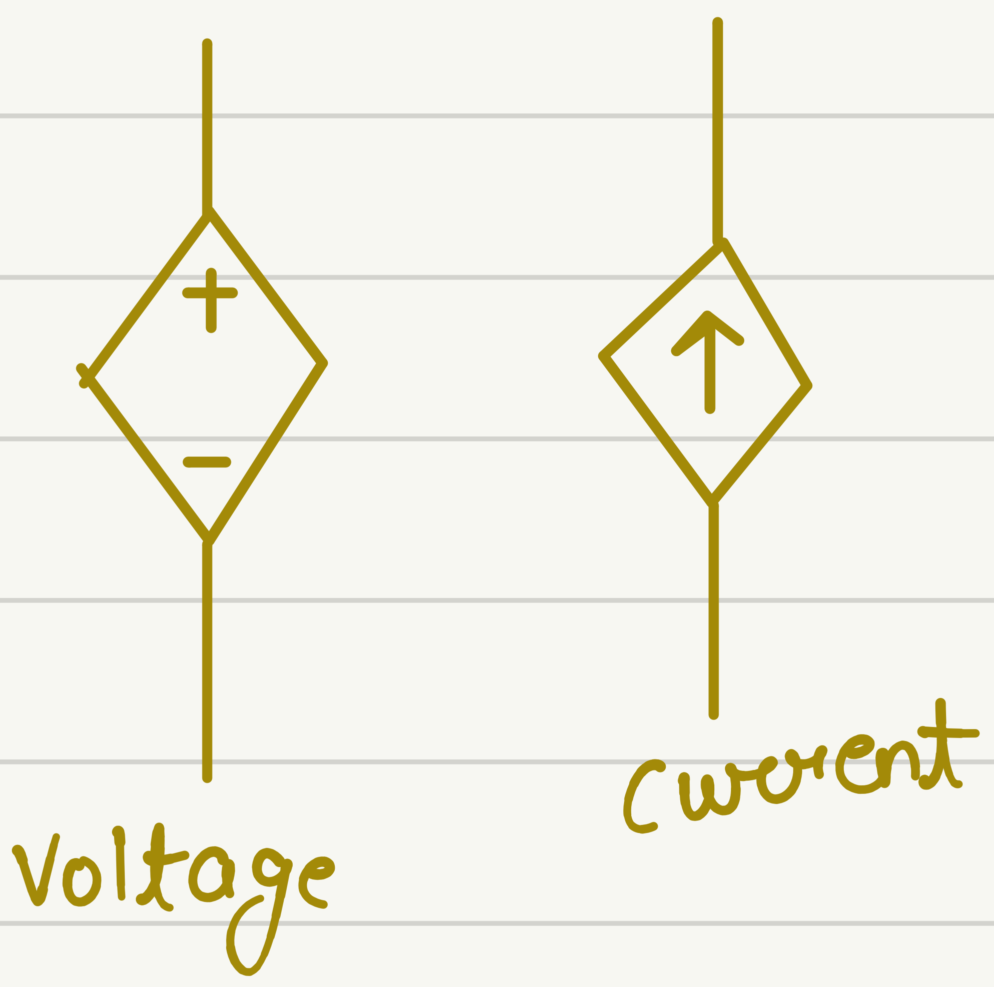
Voltage controlled voltage source

(VCVS)
VDVS

$$\boxed{V_{out} = \mu V_{in}}$$

μ ← constant value.



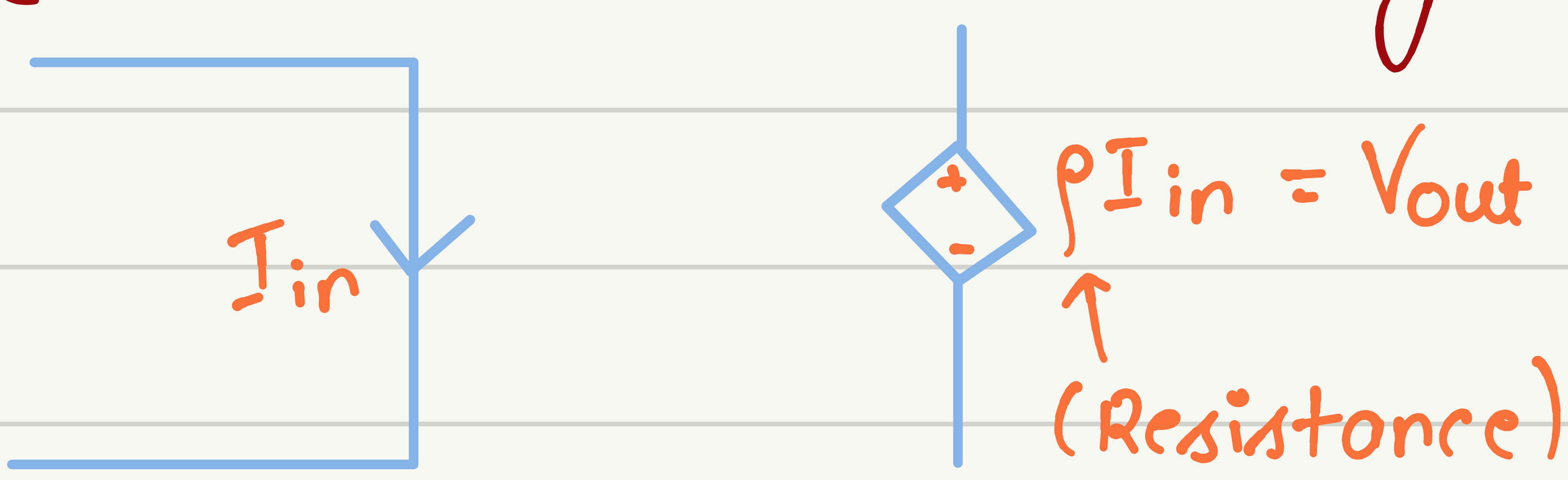


Transistor circuit

- CB
- CC
- CE

② CCVS

current controlled voltage source

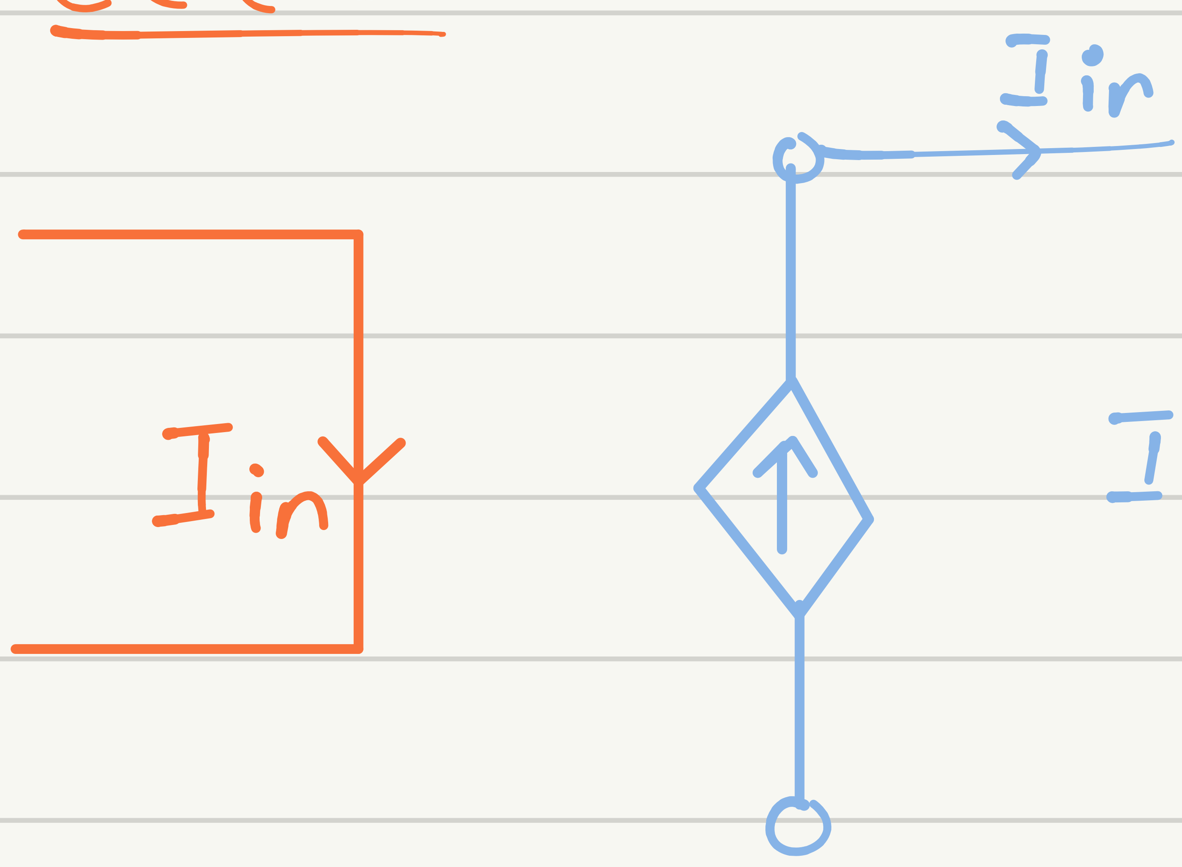


3) VCCS

$I_{out} = \alpha V_{in}$

α: $(\Omega^{-1}) \rightarrow$ constant

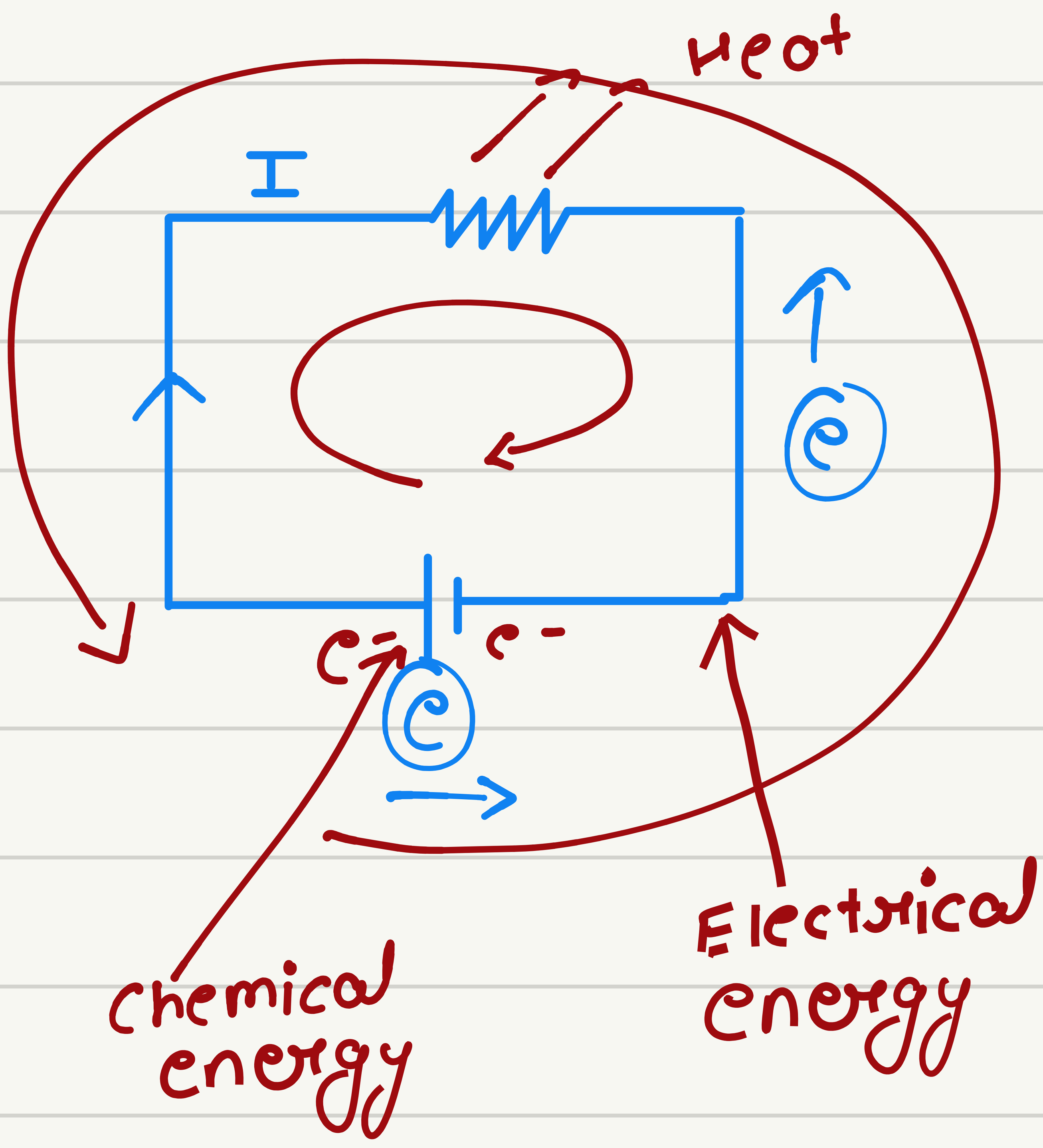
4) CCCS

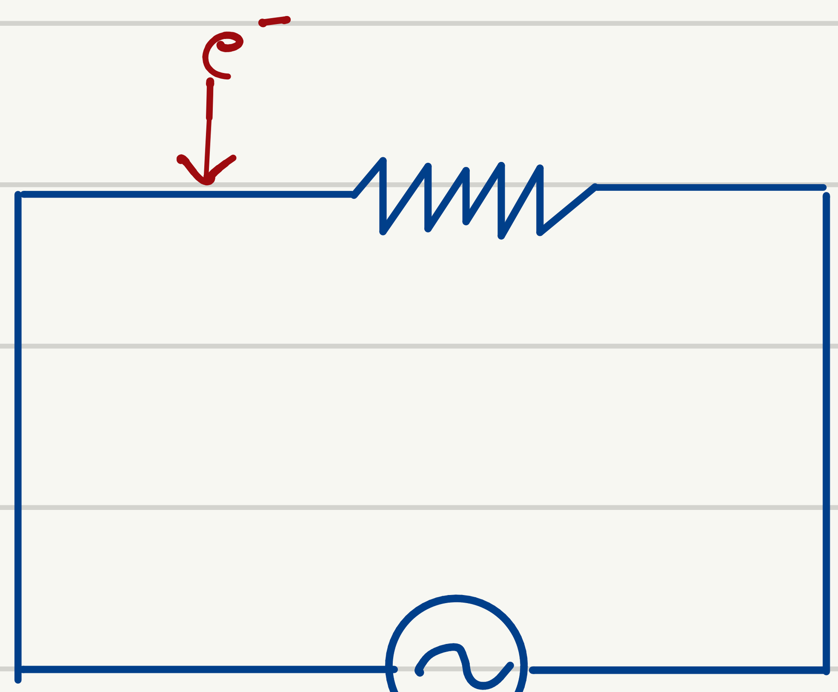


$I_{out} = \beta I_{IN}$

β: (unitless)

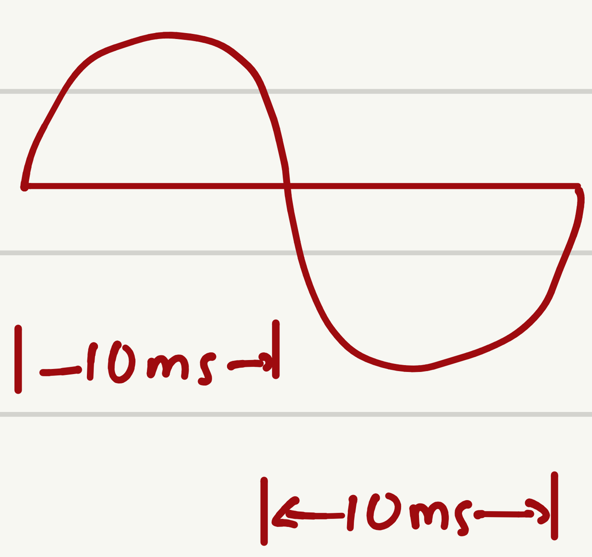
• Passive Elements





50 Hz

T = 20ms



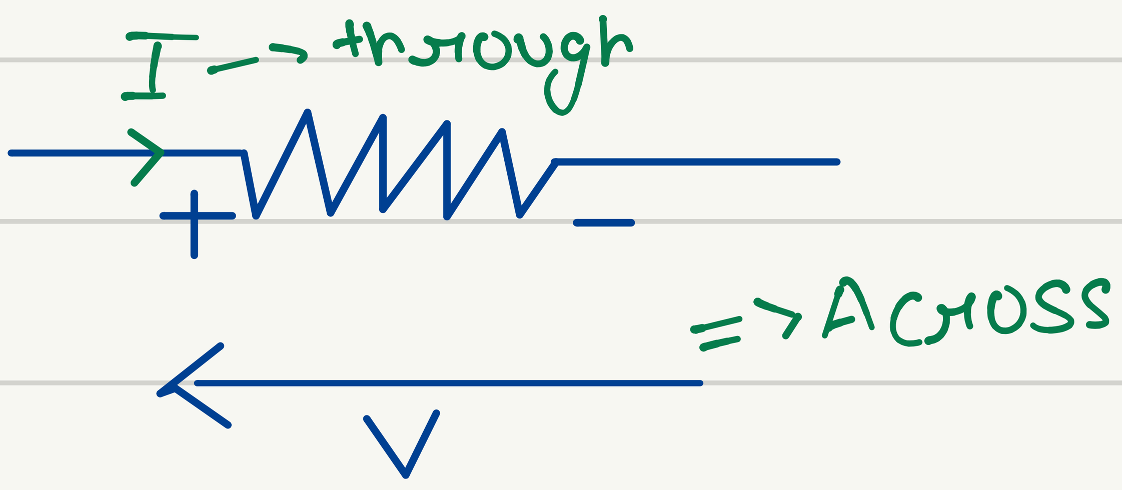
$$\boxed{V = IR}$$

Excitation $\longrightarrow V$

Response $\longrightarrow I \implies I = \frac{V}{R}$

Excitation $\longrightarrow I$

Response $\longrightarrow V \implies V = IR$



- $\underbrace{L \,\&\, C}_{\text{memory}}$

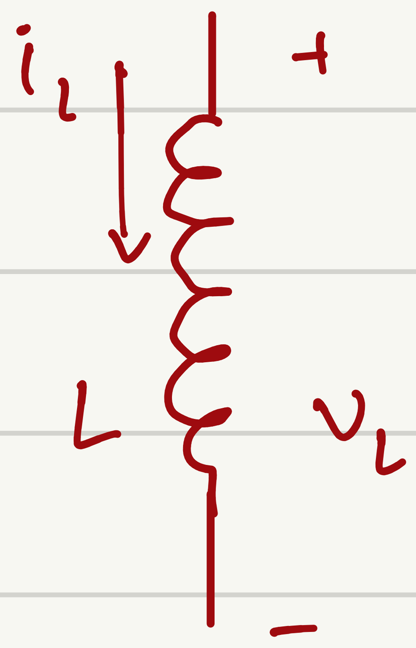
-> initial condition is important

$$\boxed{V_L = L\frac{di_L}{dt}}$$

$$\boxed{i_C = C\frac{dv_C}{dt}}$$

* Ohm's law for induction

(cause effect relationship)

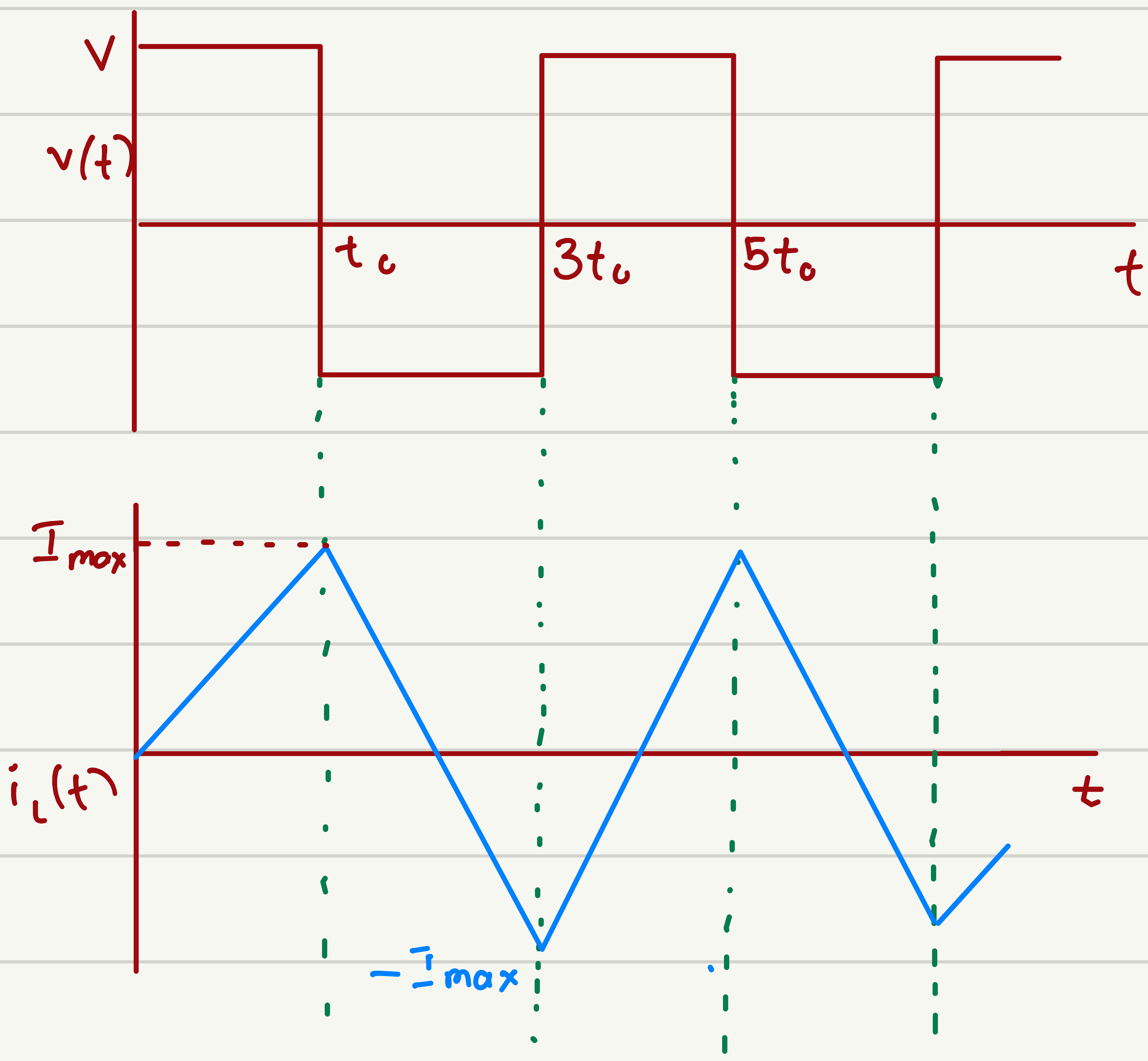


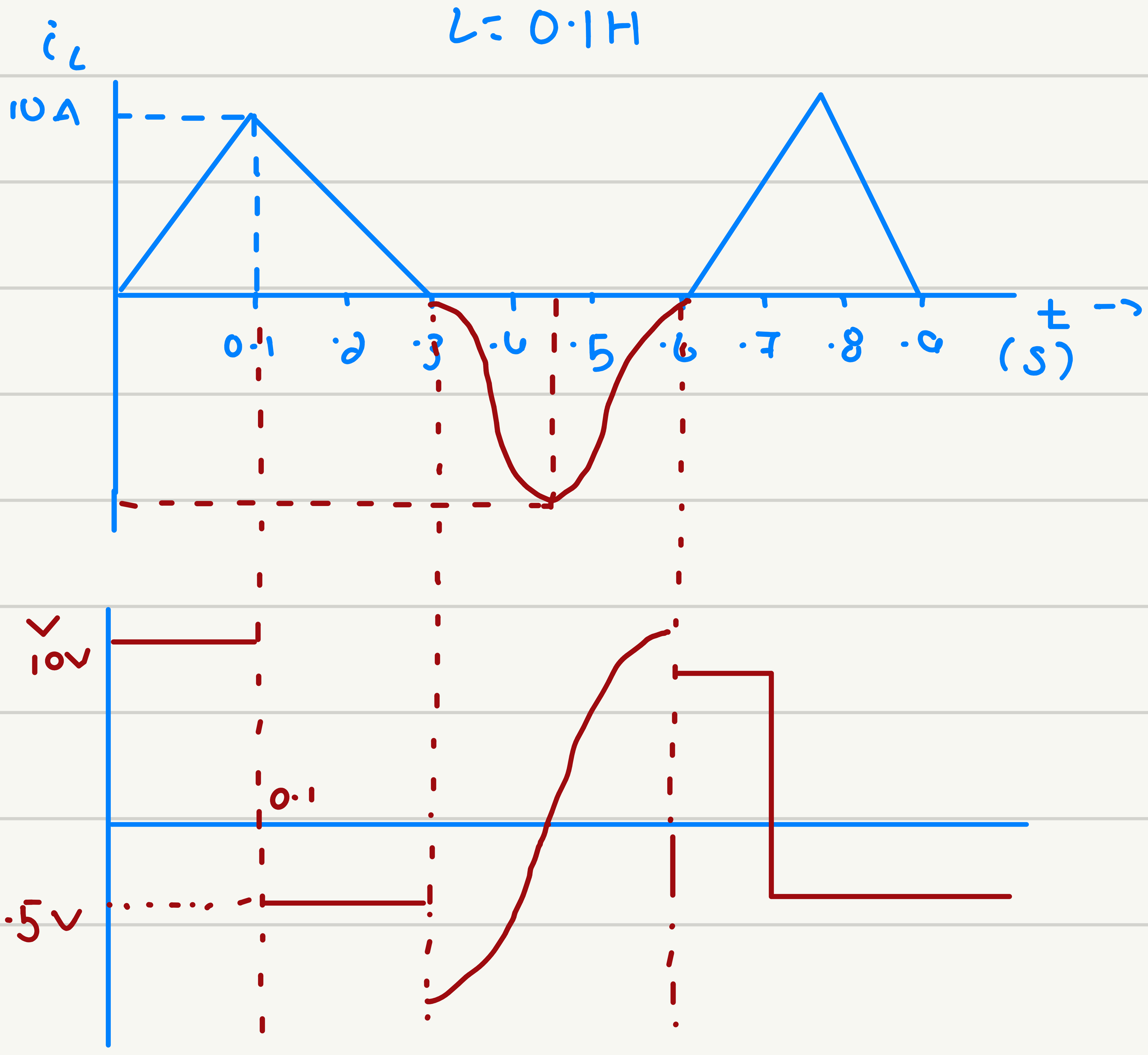
$$V_L = L\frac{di_L}{dt}$$

$V = V_L(t)$

$$i_L(t) = \frac{1}{L}\int_{-\infty}^{t} V(t)\,dt$$

$$= \frac{1}{L}\int_{t_0}^{t} v(t)\,dt + i(t_0)$$

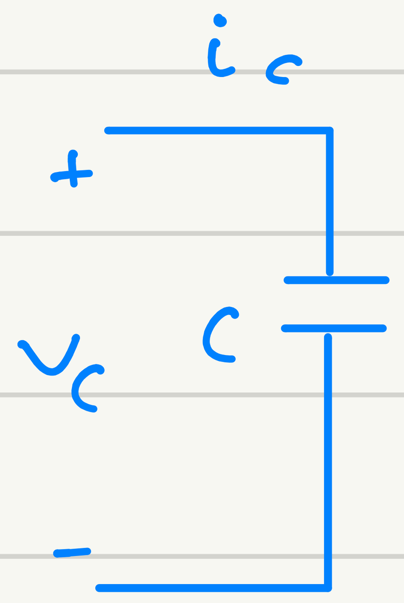




* Ohm's law for Capacitor

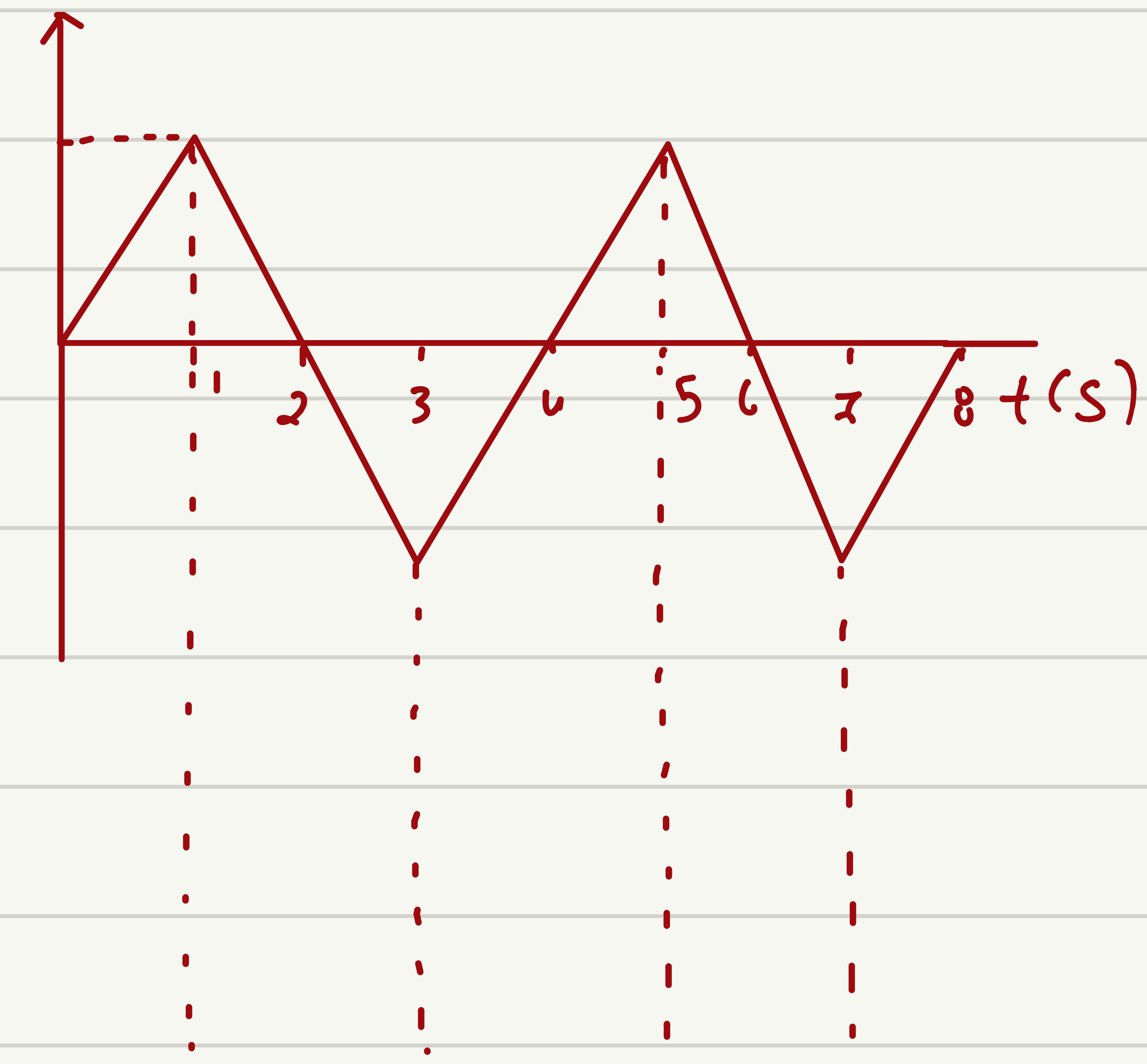
[Cause effect relationship]

$$i_C = C\frac{dv_C}{dt}$$



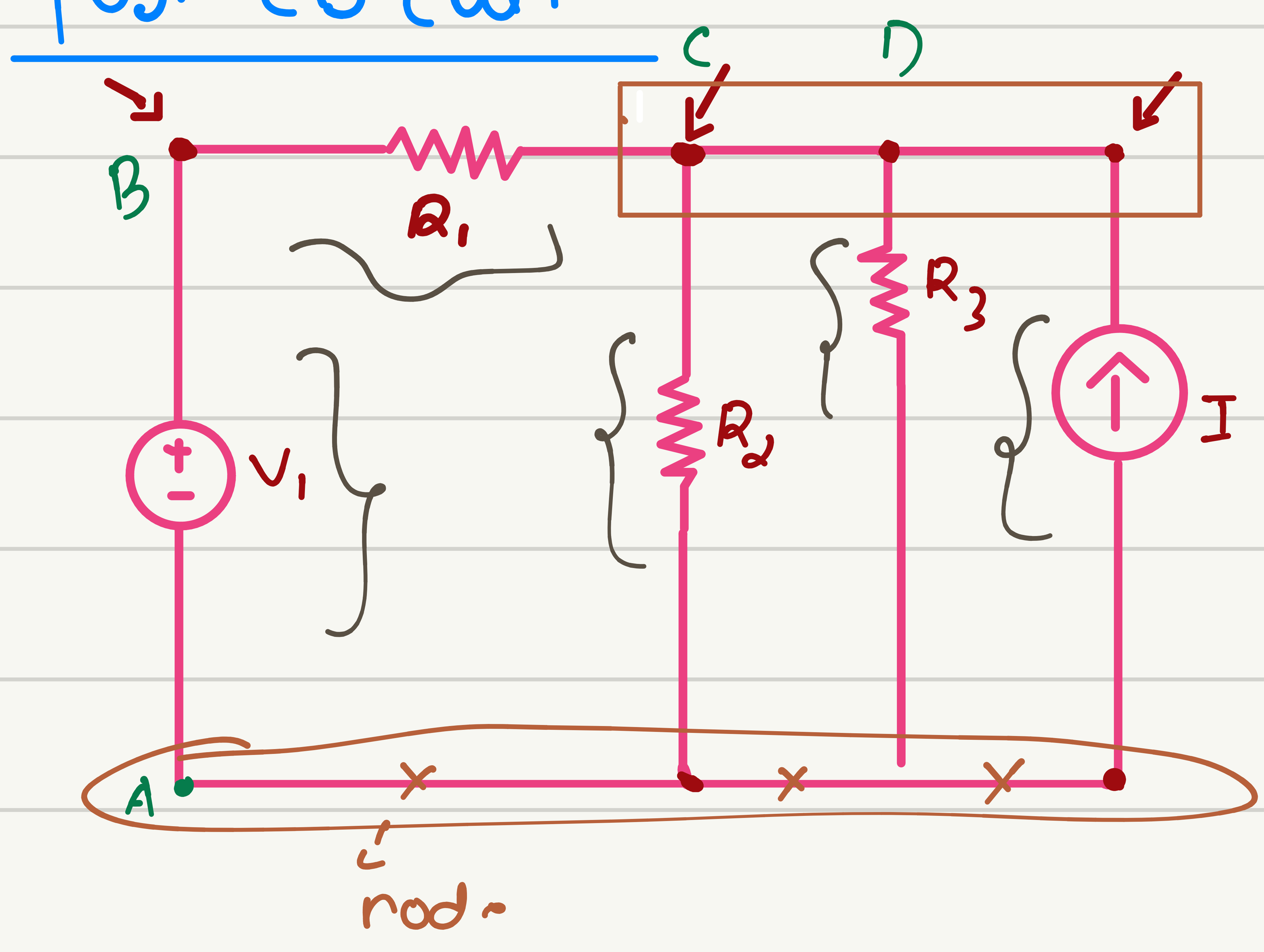
$$V_c(t) = \frac{1}{C}\int_{-\infty}^{t} i_C\,dt$$

$$V_c(t) = \frac{1}{C}\int_{t_0}^{t} i\,dt + v_c(t)$$



V → can change immediately but i cannot.

* Few important terminologies for circuit

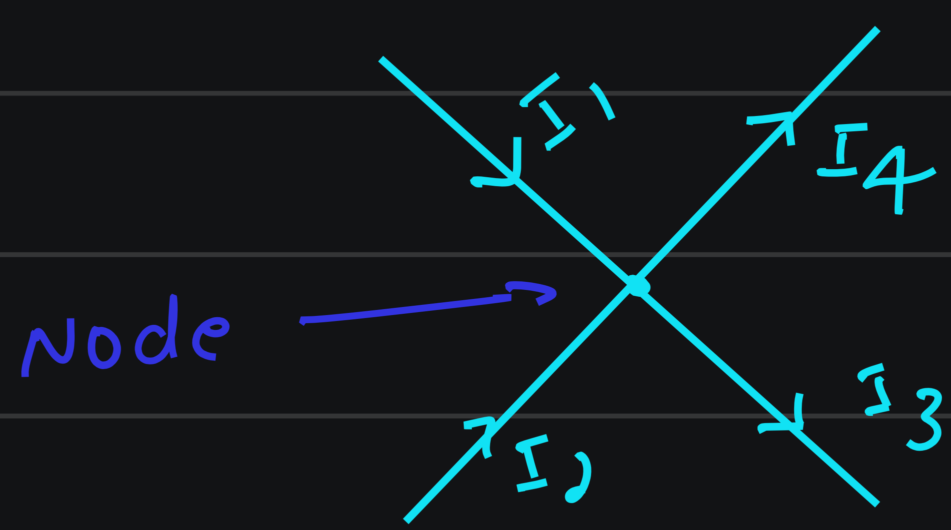


(1) Node → 3 nodes
(2) Branch → 5
(3) Loop → ABCDA
(4) Mesh → ABCA

Assign voltage to one node.

* KCL & KVL (review)

KCL (conservation charge)



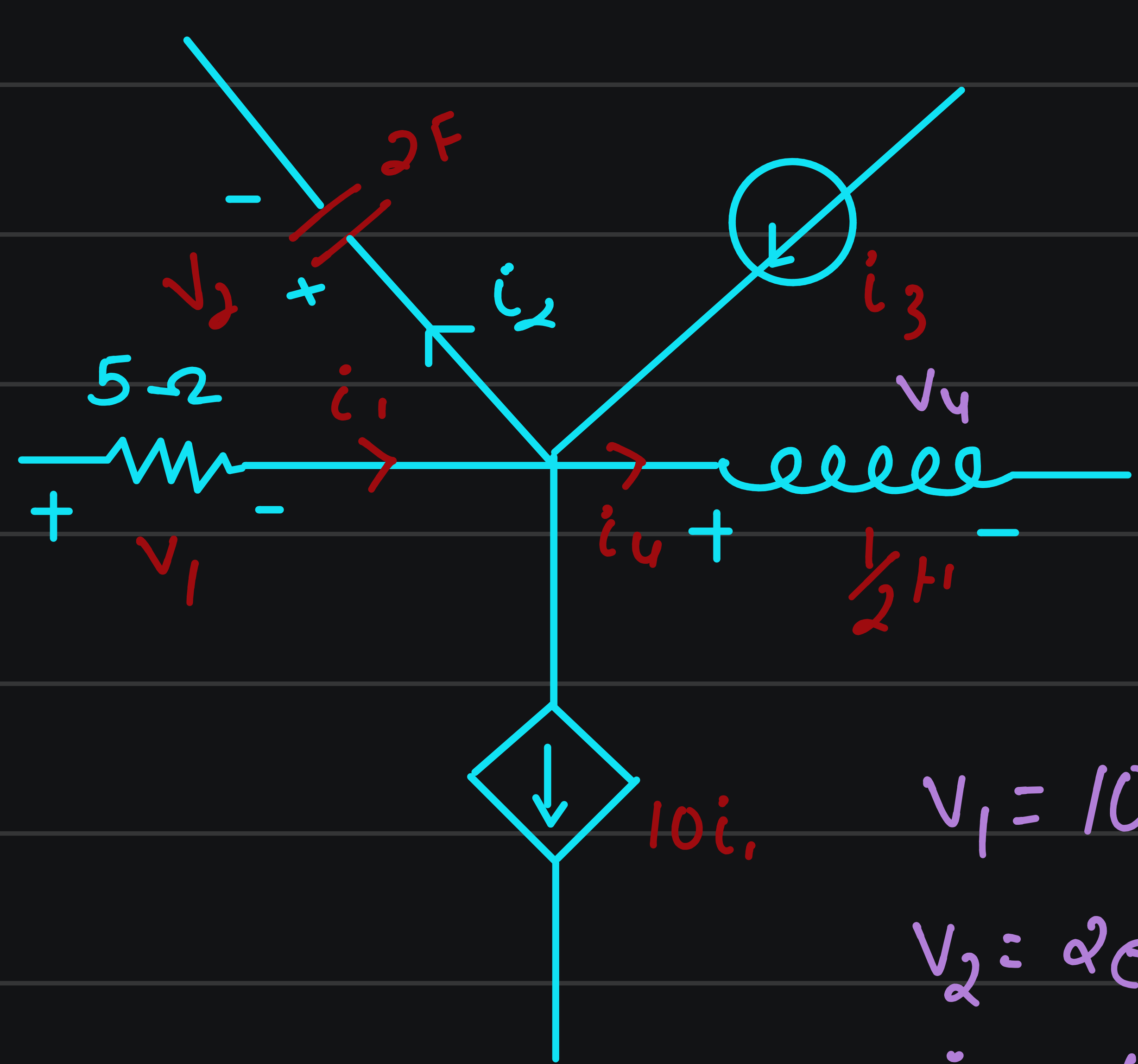
$$\sum I_i = 0$$

Entering current = Leaving current

$$\boxed{-I_1 - I_2 + I_3 + I_4 = 0}$$

Entering current -ve

Leaving current +ve.



$V_1 = 10e^{-2t}$ v

$V_2 = 2e^{-2t}$ v

$i_3 = 6e^{-2t}$ v

Find V_4.

$$-i_1 + i_2 - i_3 + i_4 + 10i_1 = 0 \quad \text{—— (1)}$$

$$= \frac{10}{5}e^{-2t} + 2.(-2)\,2e^{-2t} - 6e^{-2t}$$

$$+ i_4 + \frac{10\times 10e^{-2t}}{5} = 0$$

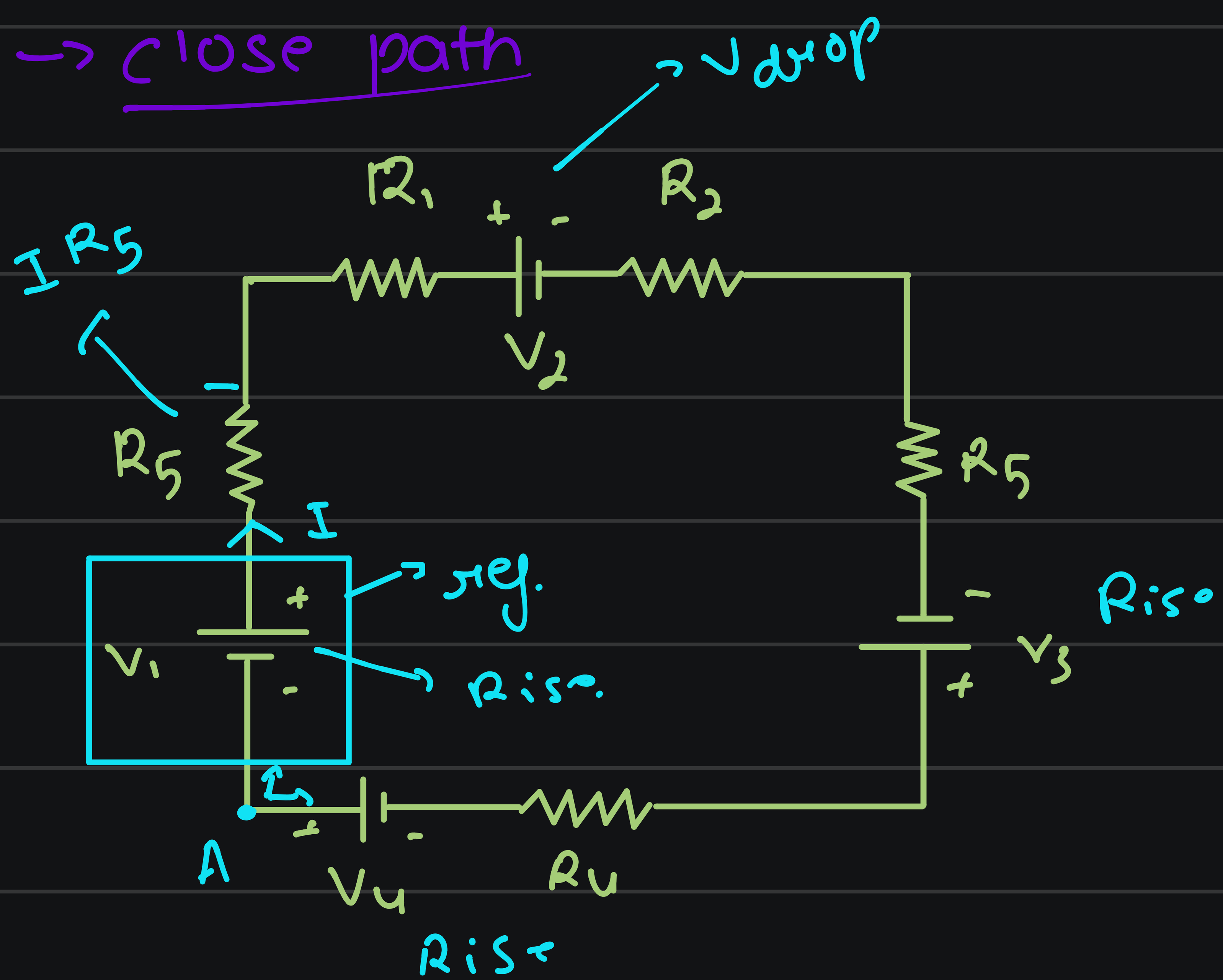
$$= i_4 = -4e^{-2t}$$

$$v_4 = L\frac{di}{dt} = 0.5\times(-4)(-2)e^{-2t}$$

$$= 4e^{-2t} \text{ volts.}$$

KVL (conservation of energy)

→ close path



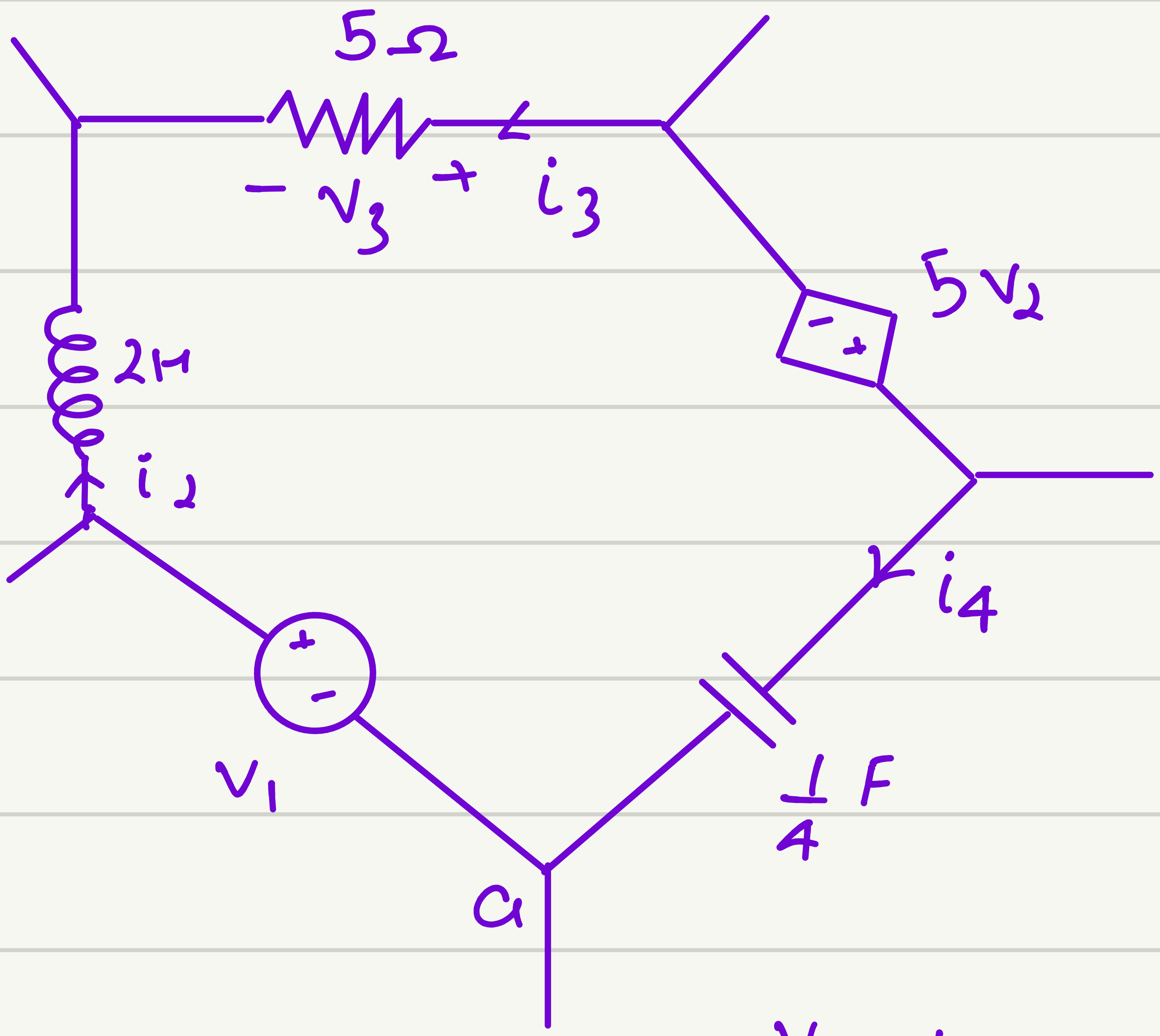
$$\sum V_i = 0$$

Voltage → with ref.

$$\sum V_{Rise} = \sum V_{drop}$$

$$\sum_{0}^{n} V_i = 0$$

$V_1 - IR_5 - IR_1 - V_2 - IR_2 - IR_3 + V_3 - IR_4 + V_4 = 0$



$V_1 = 12\sin 2t$

$i_2 = 4\sin 2t$

$i_3 = 2\sin 2t - 4\cos 2t$

$V_1 - V_2 + V_3 + 5V_2 - V_4 = 0$

$V_2 = 2 \times \frac{d}{dt}(4\sin 2t) = 16\cos 2t.$

$V_3 = 5i_3 = 5(2\sin 2t - 4\cos 2t)$

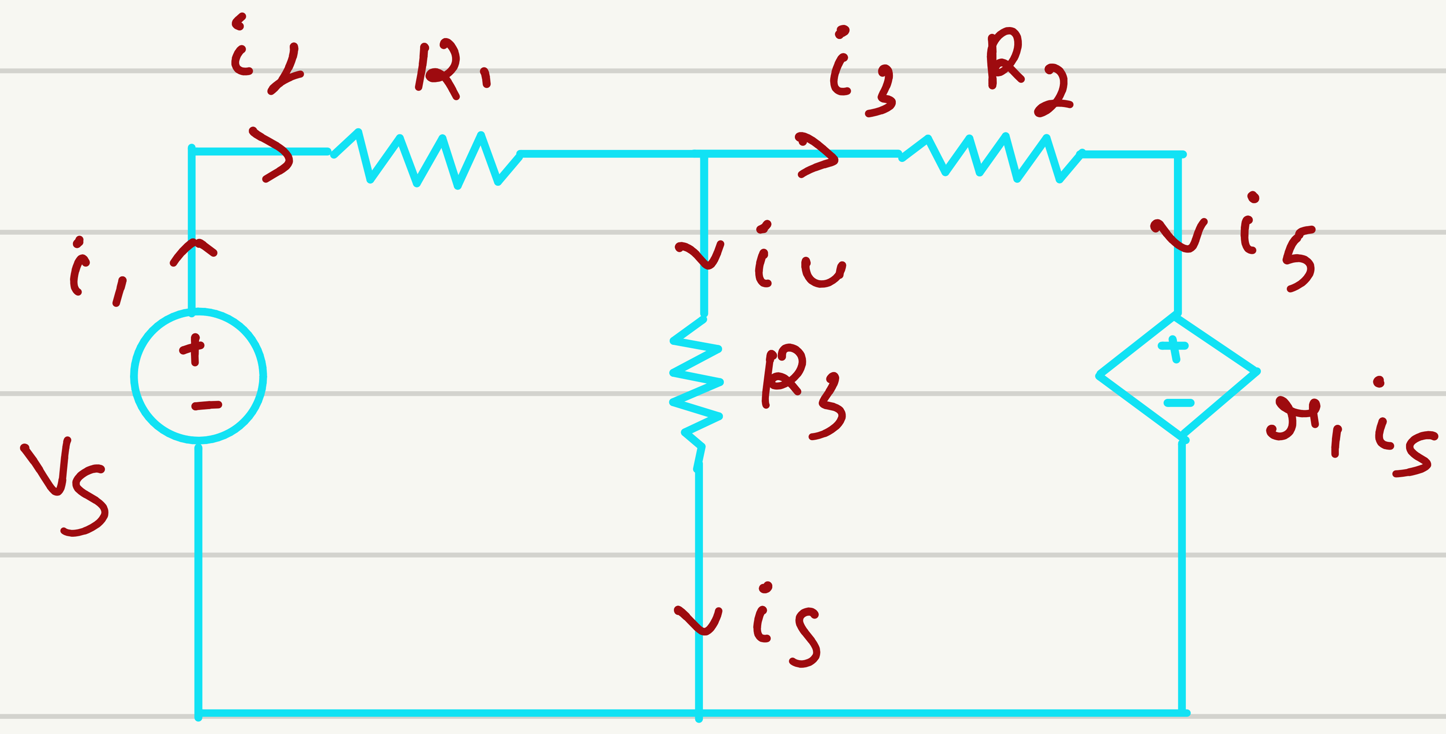
$V_4 = V_1 - V_2 + V_3 + 5V_2.$

$V_4 = 12\sin 2t - 16\cos 2t + 10\sin 2t$
$- 20\cos 2t + 80\cos 2t$
$= 22\sin 2t - 44\cos 2t$

$i_4 = 11\cos 2t - 22\sin 2t.$

Implementation/Application of KCL & KVL

1. Branch current method.
2. Mesh analysis (Loop analysis)
3. Nodal analysis
4. Supermesh analysis
5. Super nod analysis.



$-i_1 + i_2 = 0$ — (1)

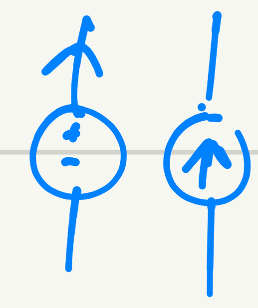
$-i_2 + i_3 + i_4 = 0$ — (2)

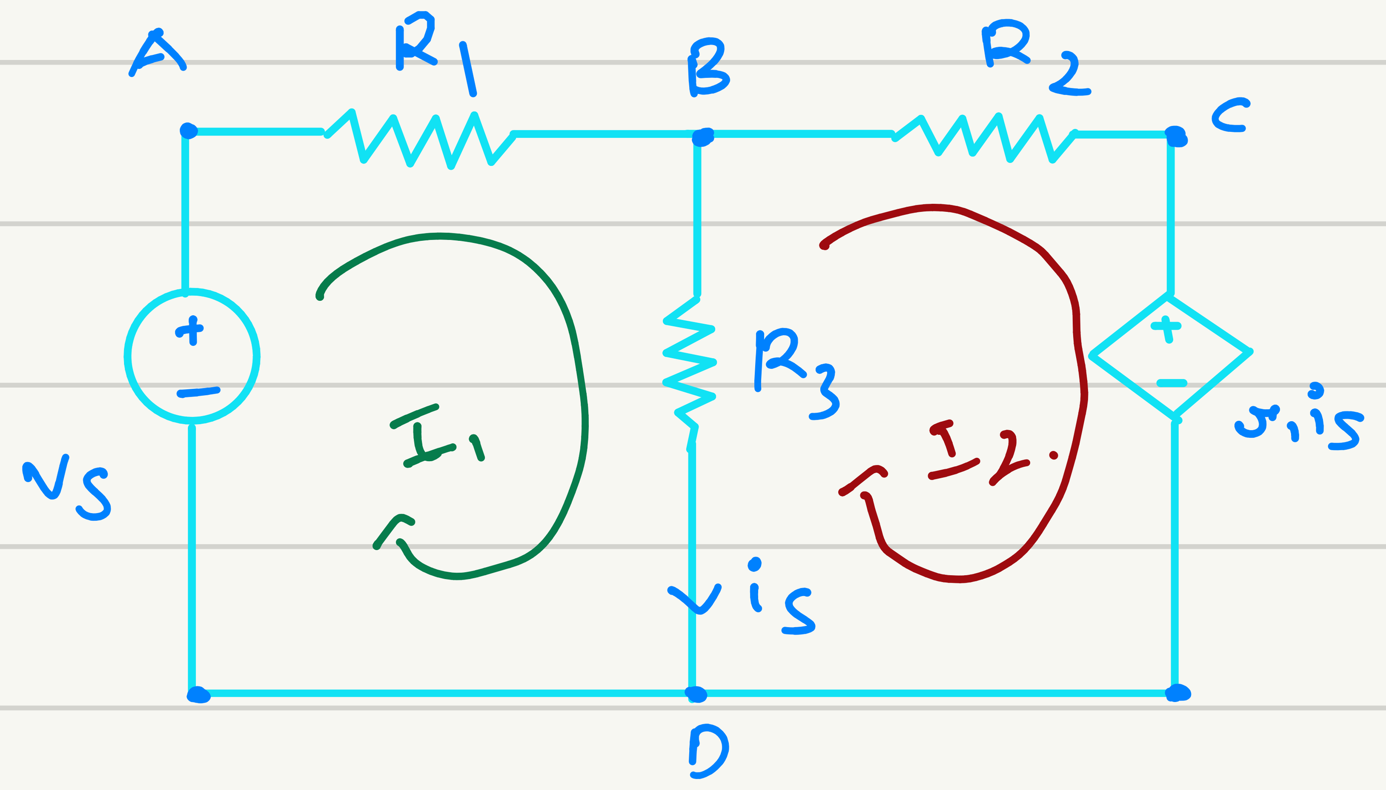
$-i_3 + i_5 = 0$ — (3)

$-i_4 + i_1 - i_5 = 0$ — (4)

$i_4 = i_5$ — (5)

→ Voltage across each branch

→ Power → dissipated (I^2R)
→ supply → VI → 
→ Absorb → VI



start

end.

Mesh₁ => ABCD

Mesh 2 => BCDB

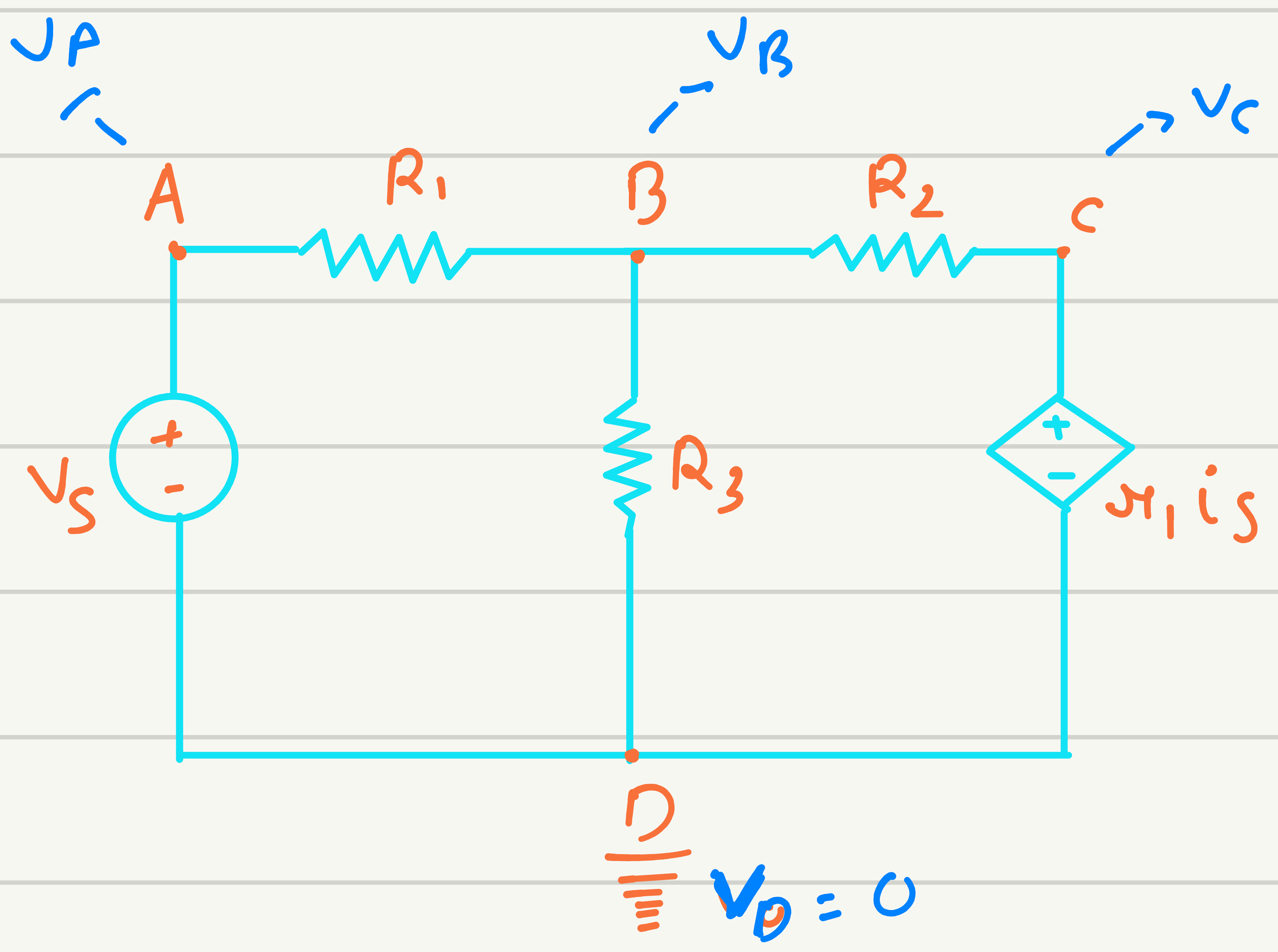
KVL eq for Mesh ABCD

$$-I_1R_1 - (I_1 - I_2)R_3 + V_S = 0 \quad \text{— (1)}$$

KVL for Mesh BCDB

$$-I_2R_2 - r_1 i_s - (I_2 - I_1)R_3 \quad \text{— (2)}$$

Casio fx 991 series calculator



Step-1: Identify nodes

A B C D

Step 2: Choose one node as Reference

→ Max branch connected

→ The lower one in picture.

Step 3 - Assign voltage to ref node

0 -25 +25 . . .

Step 4: Write KCL for each non ref node

(1) for node A

$$\frac{V_S - V_B}{R_1}$$

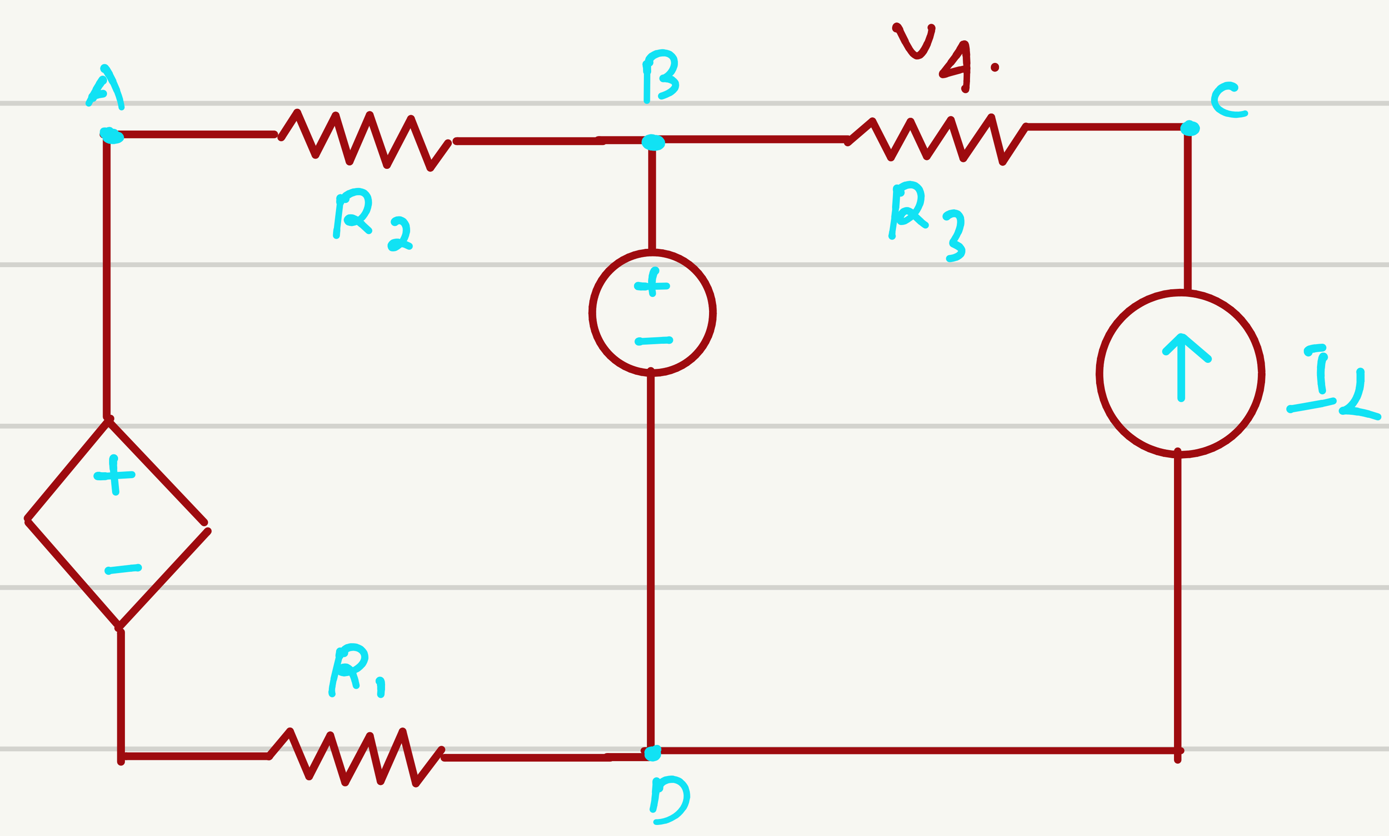
× not required

(2) for node B

$$\frac{V_B - V_A}{R_1} + \frac{V_B - V_C}{R_2} + \frac{V_B}{R_3} = 0.$$

(3) $\frac{V_C - V_B}{R_2}$ +

Not required.



Calculate current, voltage & Power in each element of the circuit with appropriate comment

Using branch current, nodal voltage analysis, mesh voltage. Try to compare.

$R_1 = 2\,\Omega$

$R_2 = 4\,\Omega$

$R_3 = 7\,\Omega$

$V_0 = 12\,V$

$I_0 = 1A$

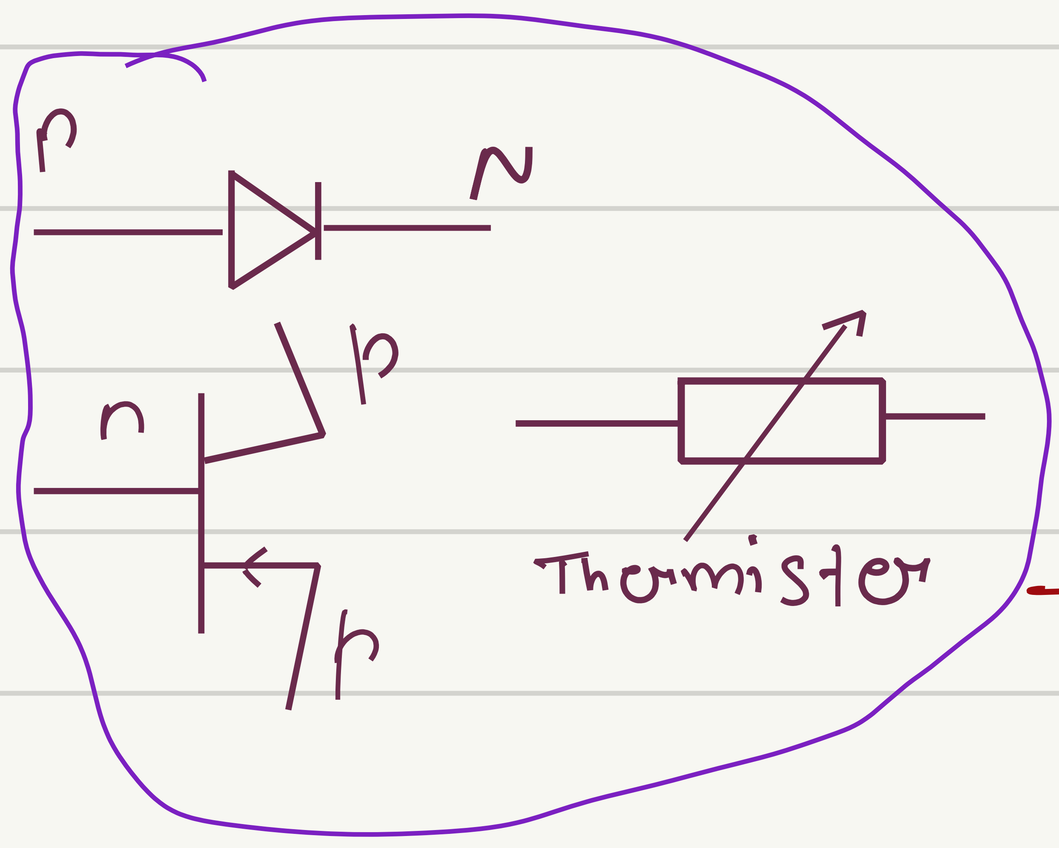
$\alpha = 0.45$

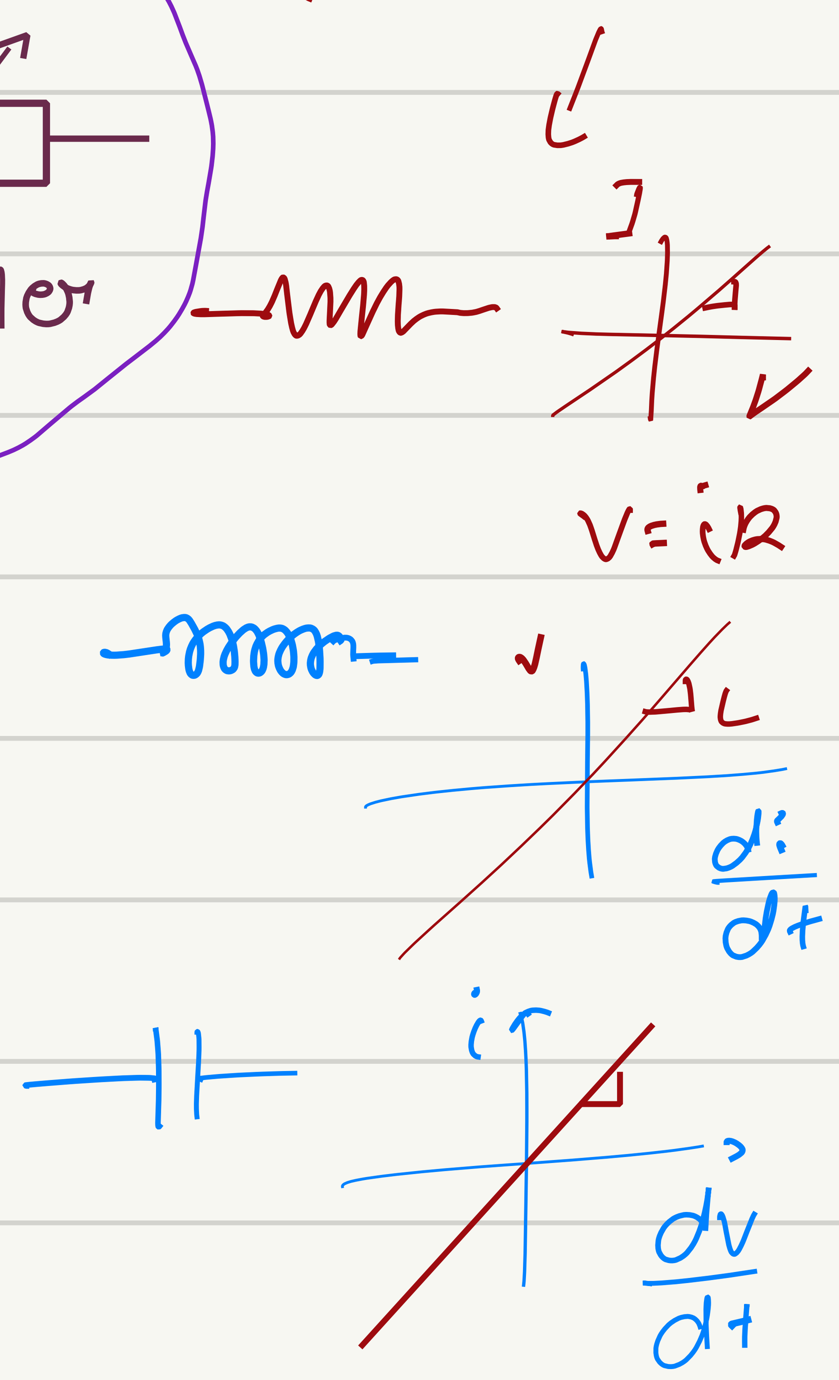
Network Theorem

↳ Partial solution of network

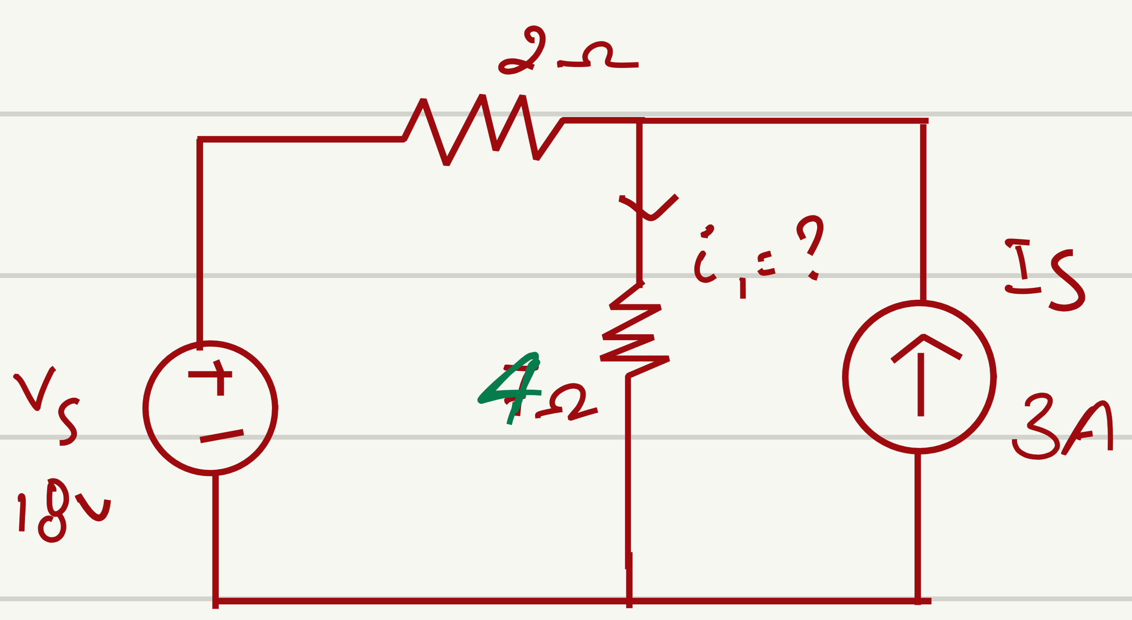
1. Super position theorem
2. The venin's theorem
3. Norton's theorem
4. Maximum Power theorem
5. Telegen's theorem

For linear network

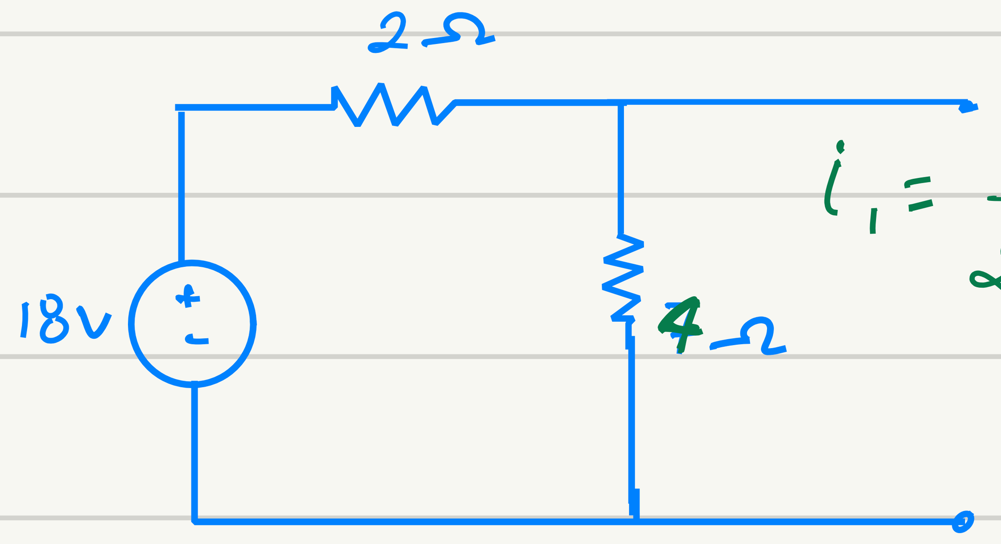




→ Super position theorem

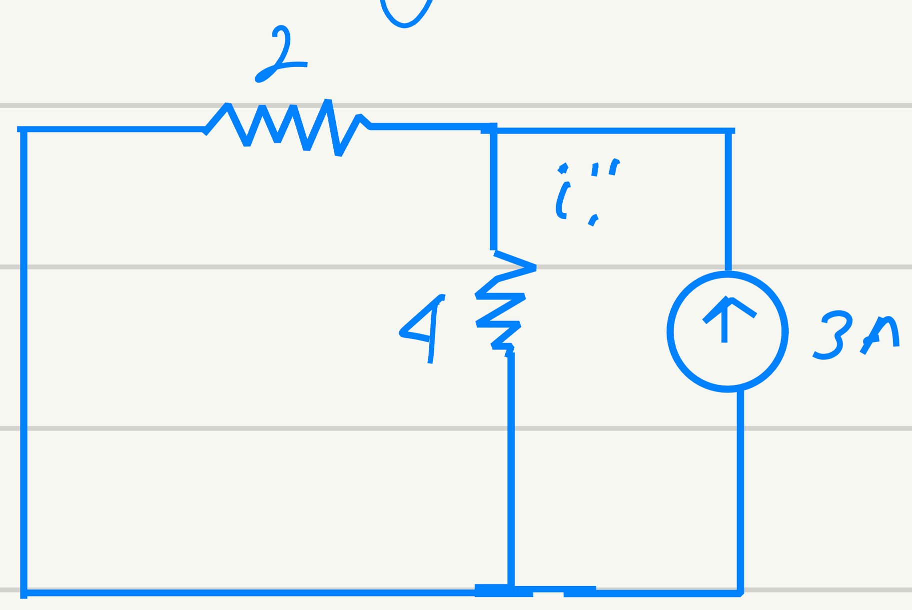


① When only 18V(Vs) is considered.



$i_1 = \frac{18}{2+4} = \frac{18}{6} = 3A$

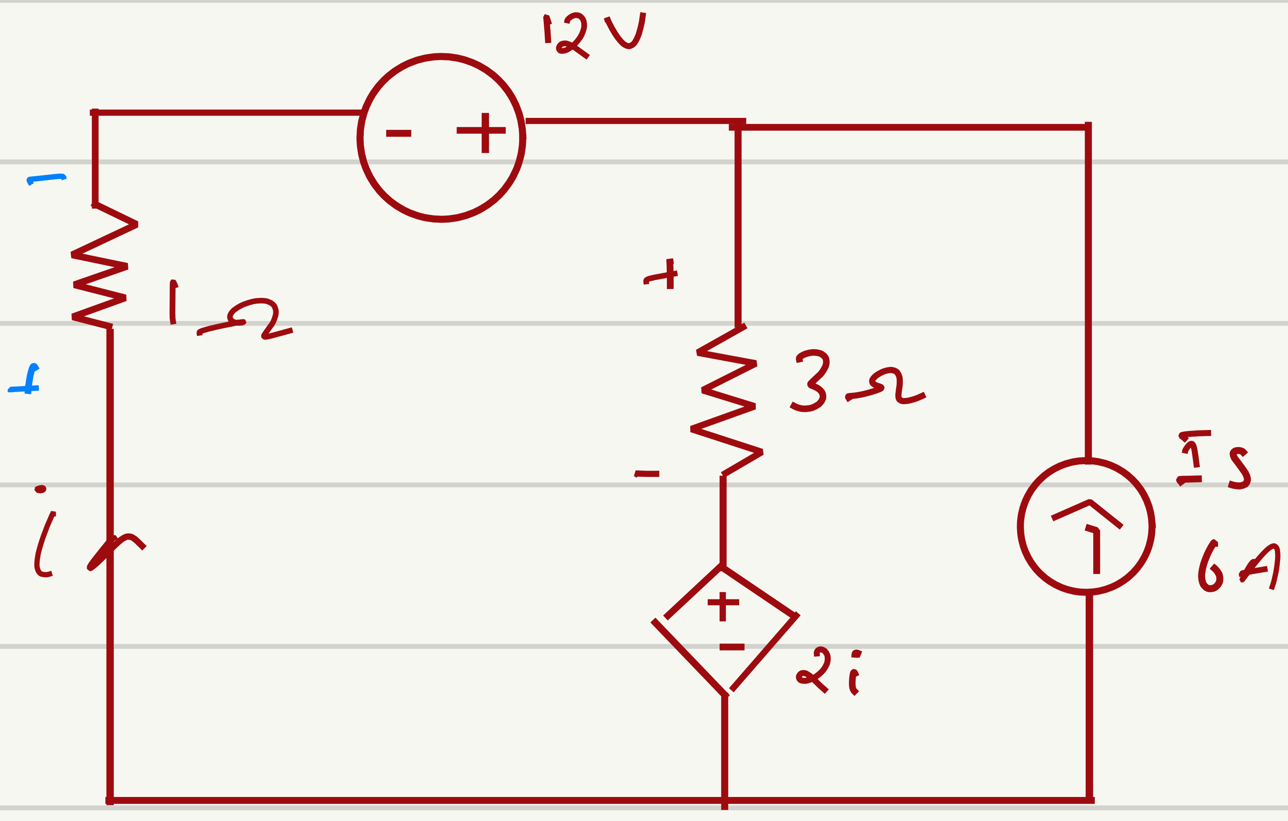
② When only 3A(I_S) is considered



$i_1'' = \frac{3}{2+4} \times 2 = 1A.$

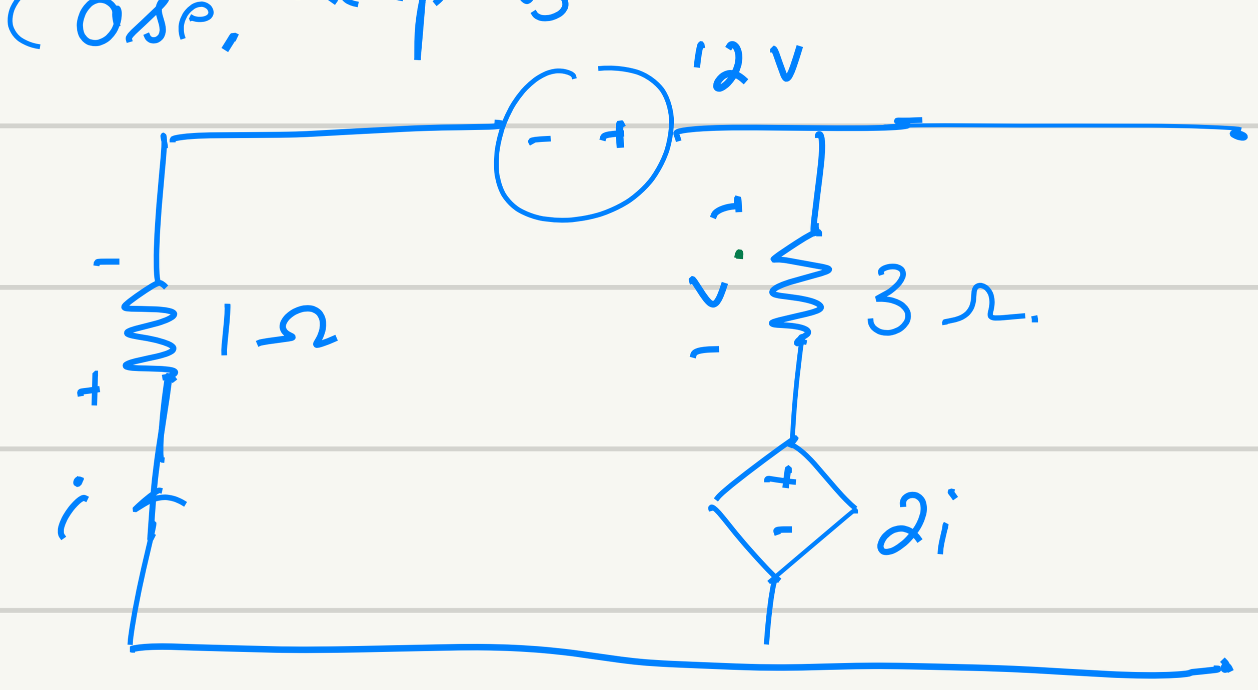
Total current

$i_1 = i_1' + i_1'' = 3+1 = 4A$



Find v

Case, keep Vs



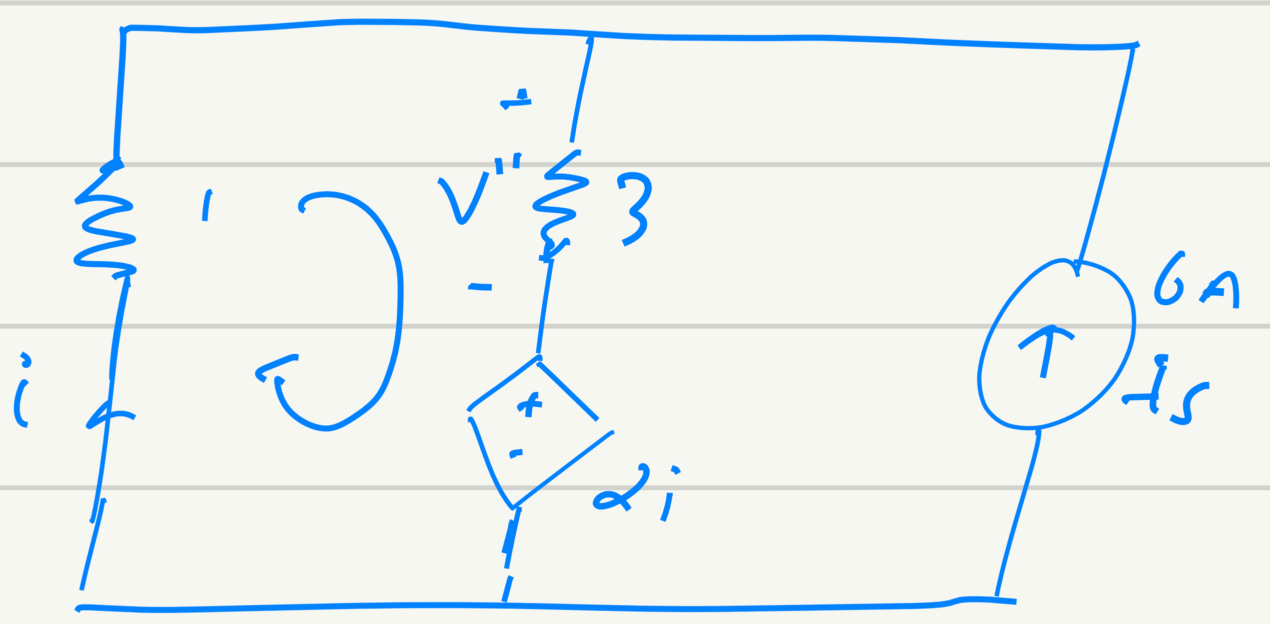
$-i + 12 - 3i - 2i = 0$

$6i = 12$

$i = 2A.$

$v' = 3 \times 2 = 6V.$

→ Case 2 - Keep I_s only

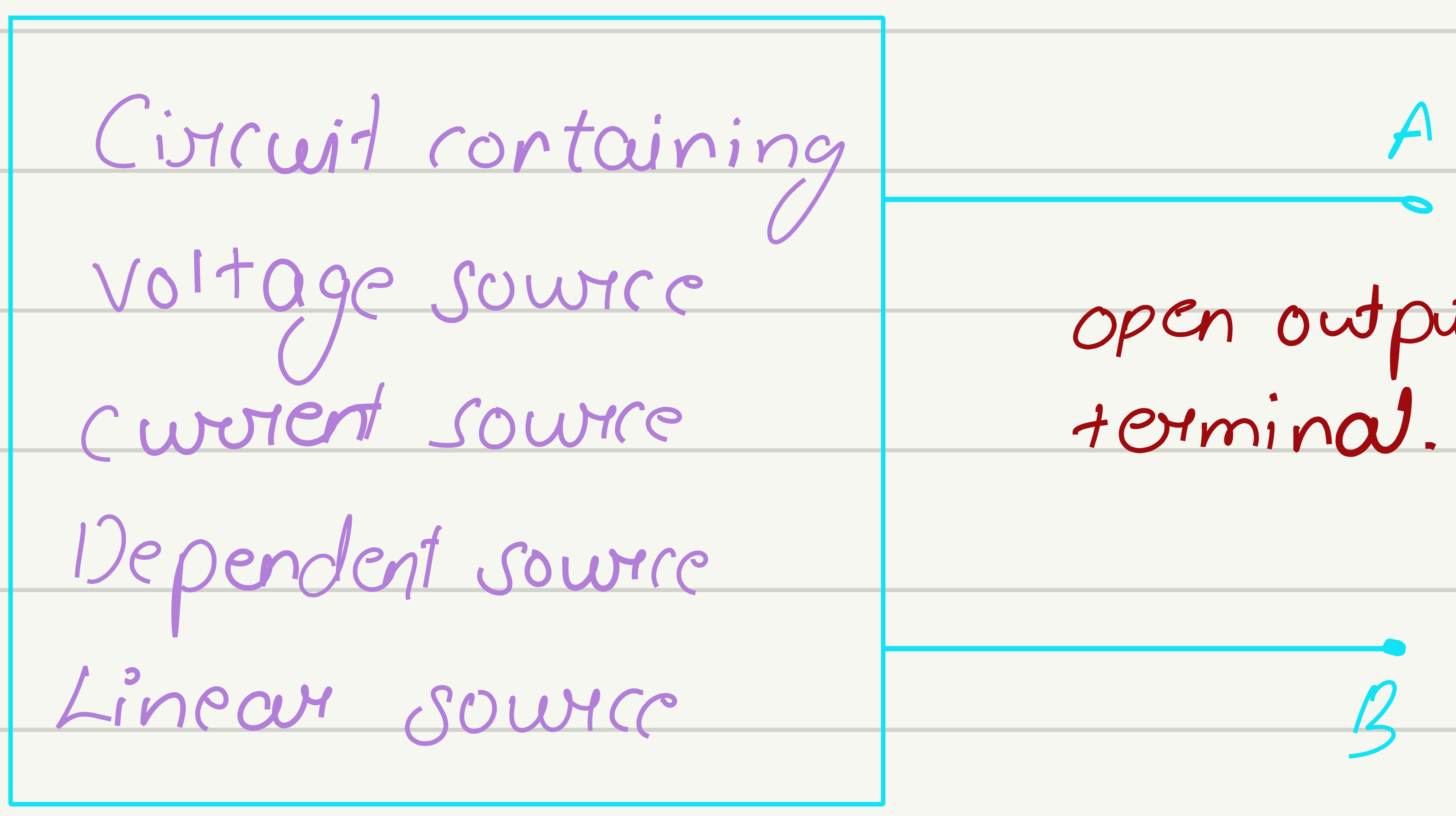


$-i - 3(i+6) - 2i = 0$

$-i - 3i - 18 - 2i = 0$

$i = 3A$

$V = (-3+6) \times 3 = 9V$



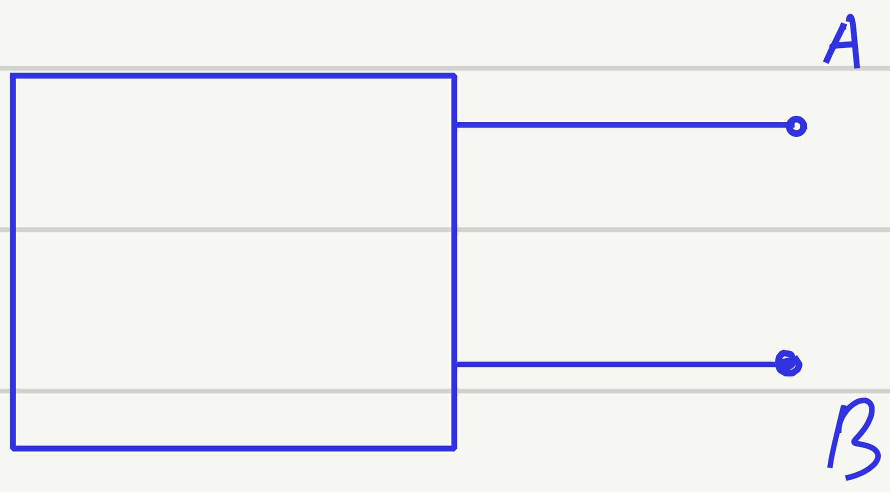
open output terminal.

To determine V_{Th}

Calculate $V_{OC} = V_A - V_B = V_{AB}$

To determine R_{Th}

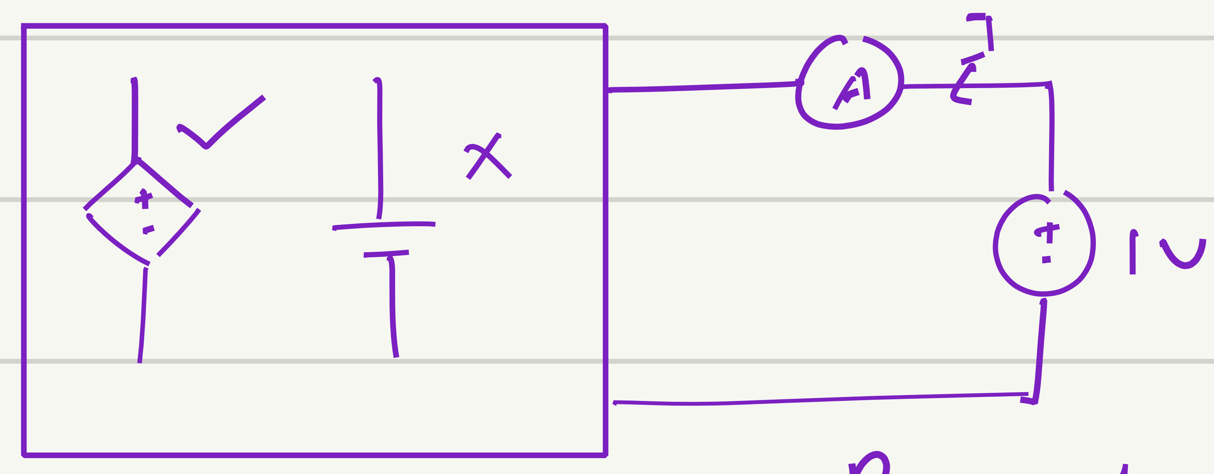
① When no dependent source



V → SC

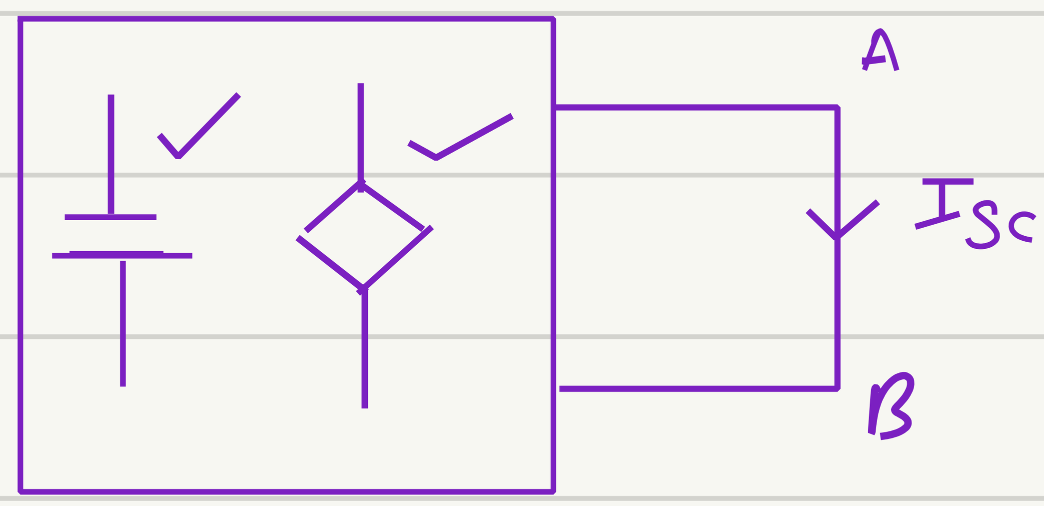
I → OC

② Using test source



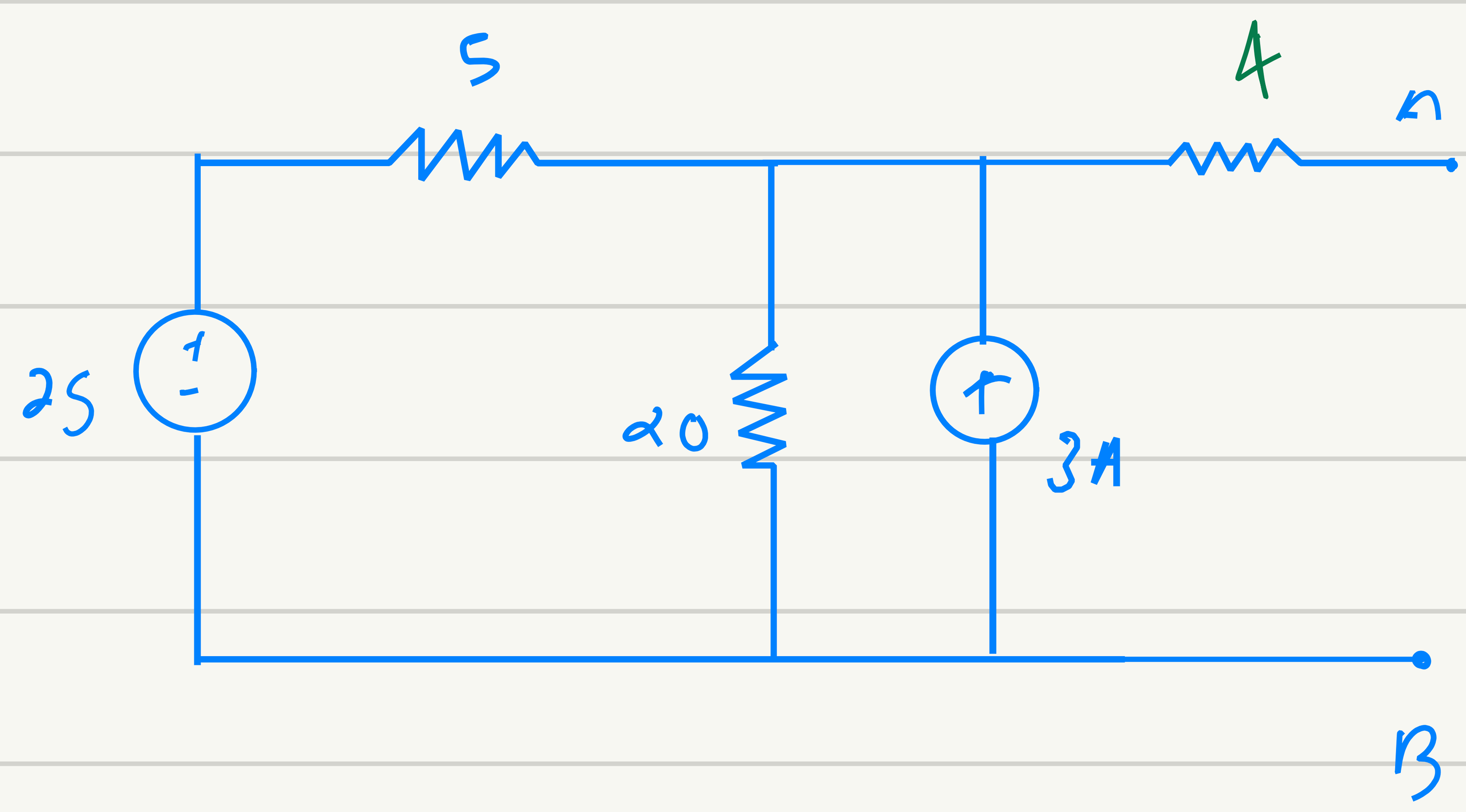
$R_{Th} = \frac{1}{I}$

③ By short circuit current



$R_{Th} = \frac{V_{OC}}{I_{SC}}$

Q-



Find thevenin's Equivalent.

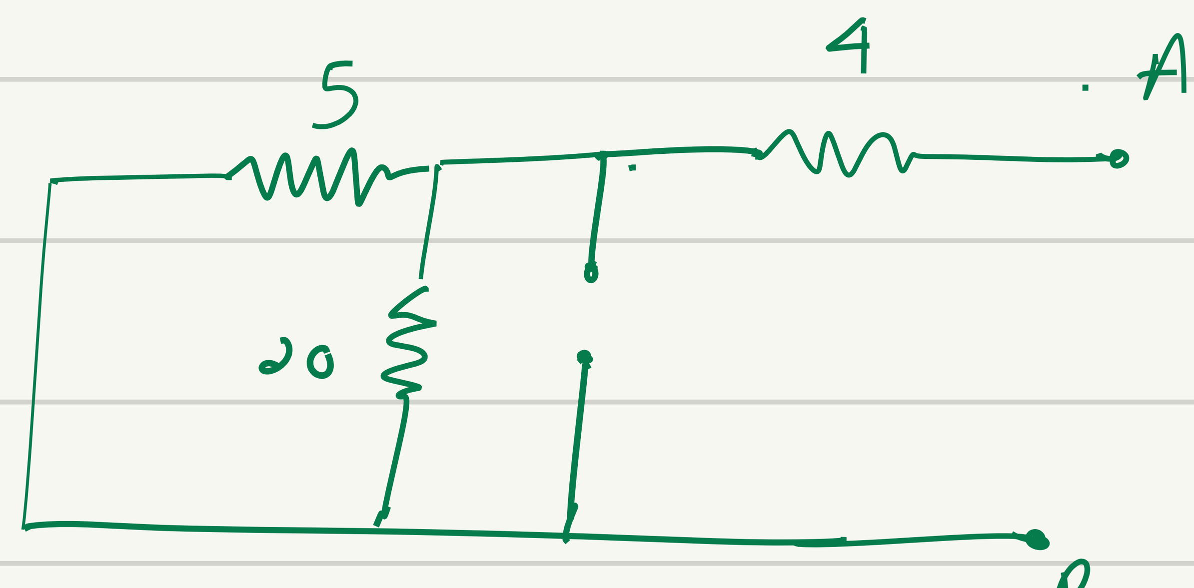
Solⁿ $V_{Th} = ?$: $V_{OC} = V_{AB} = V_{CB}$

$$\frac{V_c - 25}{5} + \frac{V_c}{20} - 3 = 0$$

$V_c = 32V = V_{CB} = V_{AB} = V_{Th}$

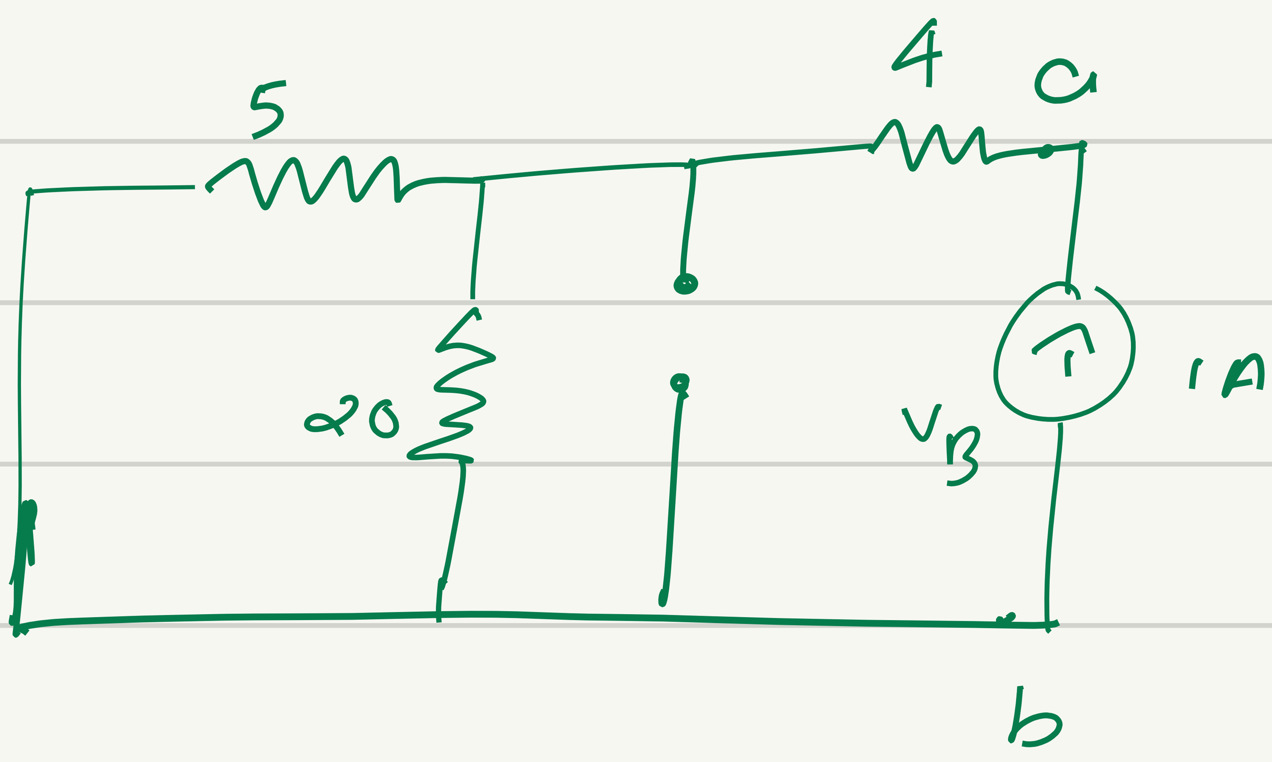
R_{Th}

M_1 -



$$\frac{5\times 20}{5+20} + 4 = 8\,\Omega.$$

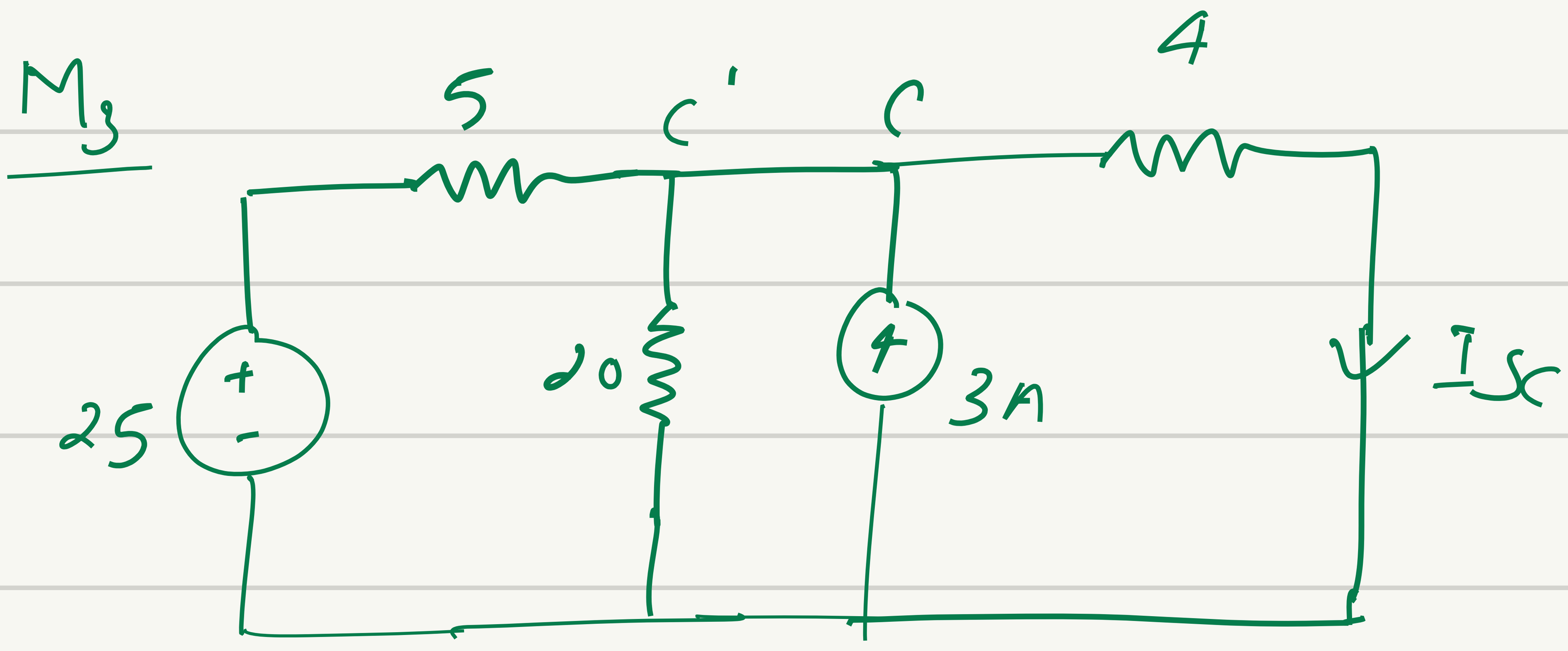
M_2 –



$V_{AB} = 8V$

$R_{Th} = 8 = 8\,\Omega.$

M_3

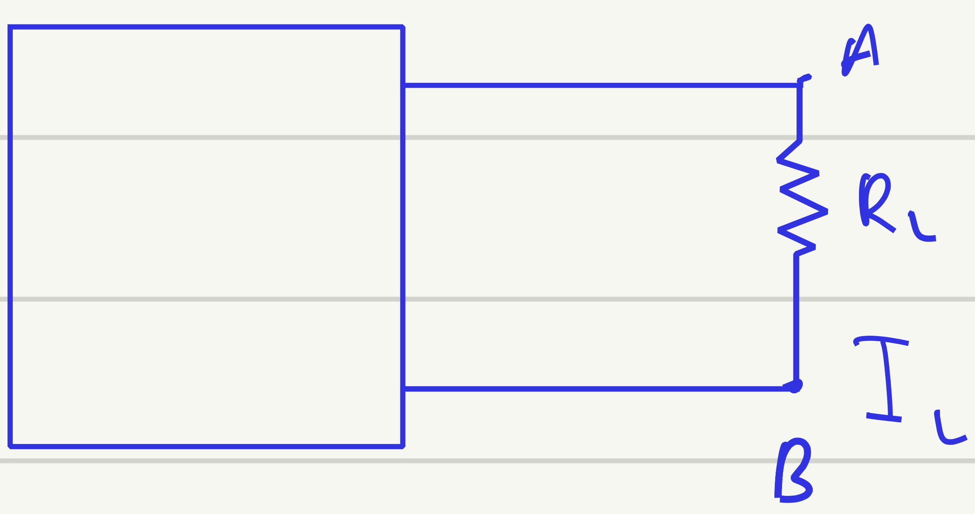


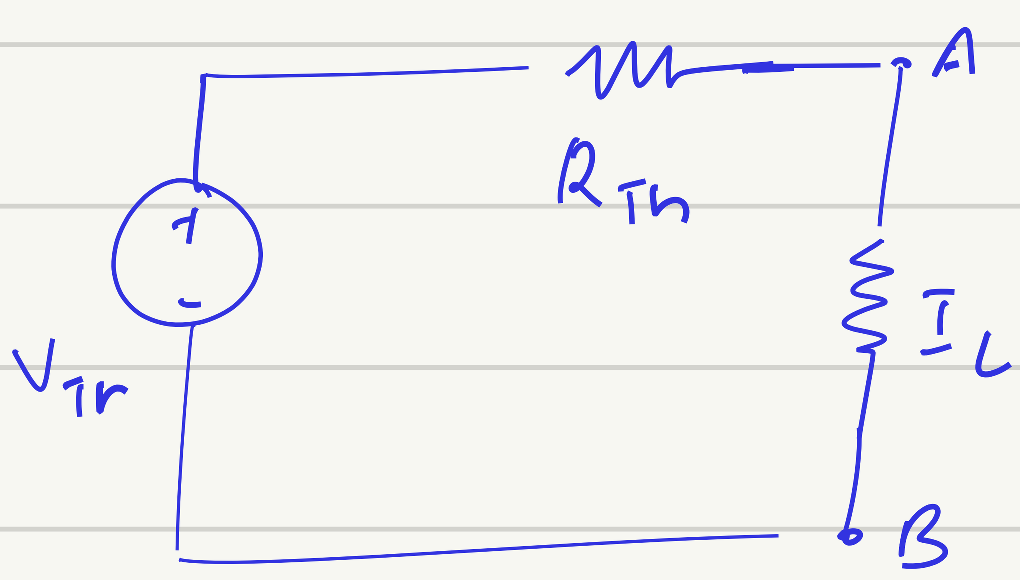
$$\frac{V_c - 25}{5} + \frac{V_c}{20} - 3 - I_{sc} = 0$$

$$\frac{V_c - 25}{5} + \frac{V_c}{20} - \frac{V_c}{4} - 3 = 0$$

$I_{sc} = 4A.$

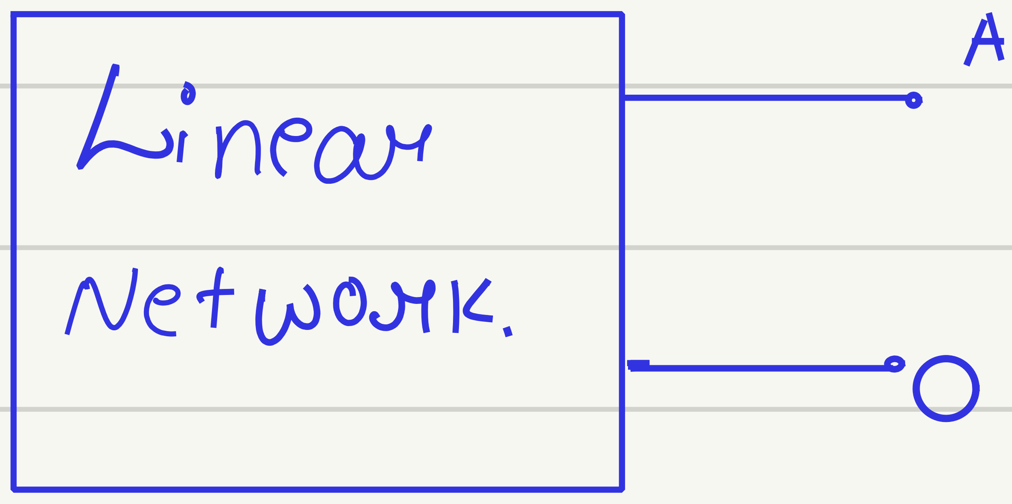
Application of Thevenin's theorem





$$I_L = \frac{V_{Th}}{R_L + R_{Th}}$$

Norton Theorem



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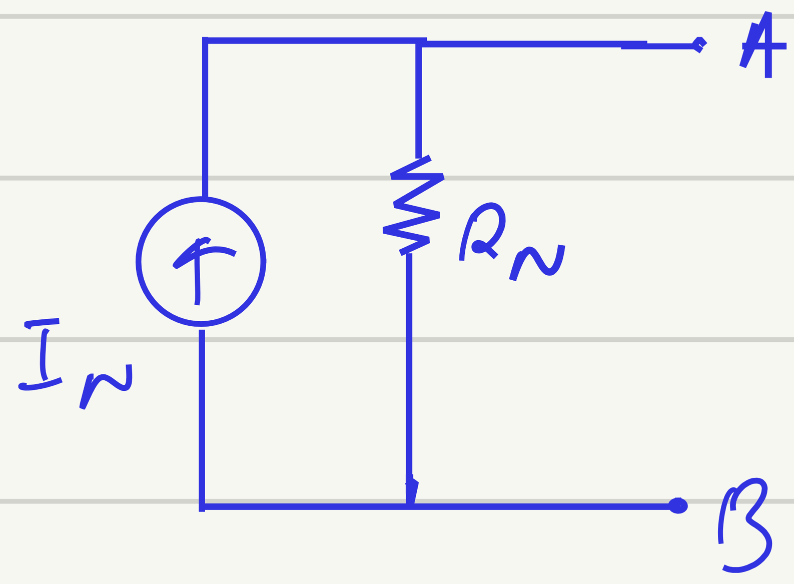


Norton equivalent circuit.

$I_N = I_S = I_{AB}$

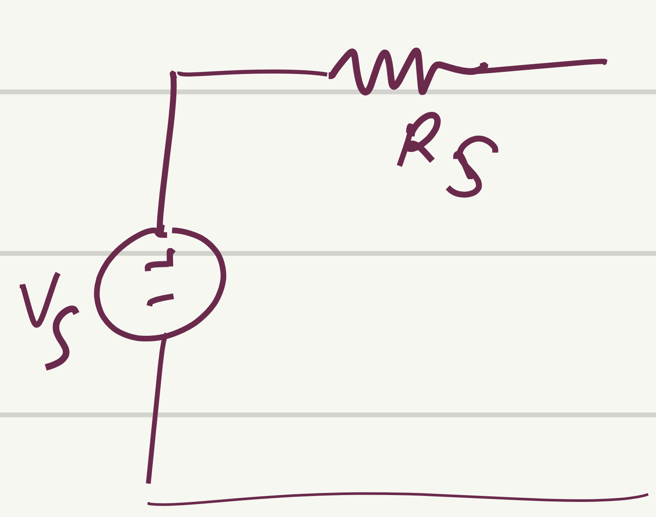
$R_N = R_{Th}$

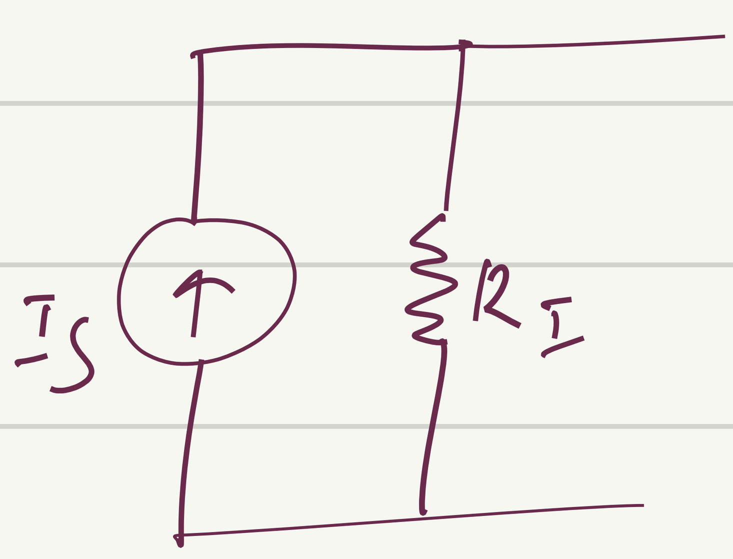
Source Transformation

Independent practical voltage source

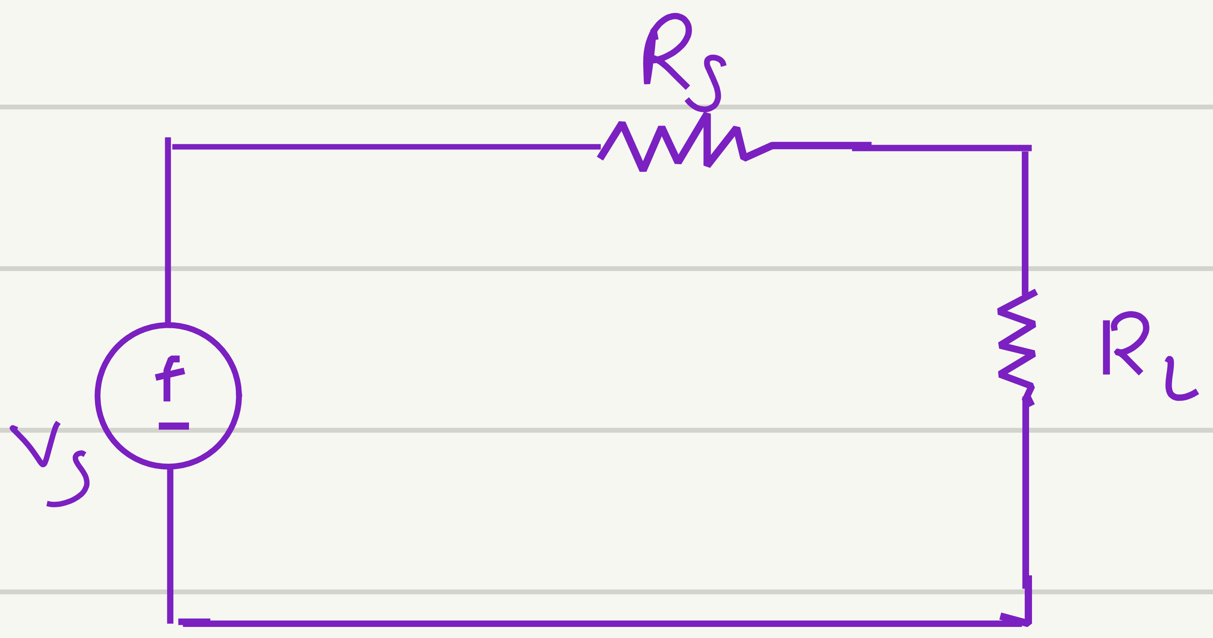
⇕

Independent practical current source.





Maximum Power transfer theorem



P_L max when $R_S = R_L$

Tellegen's Theorem

(Linear/non-linear network)

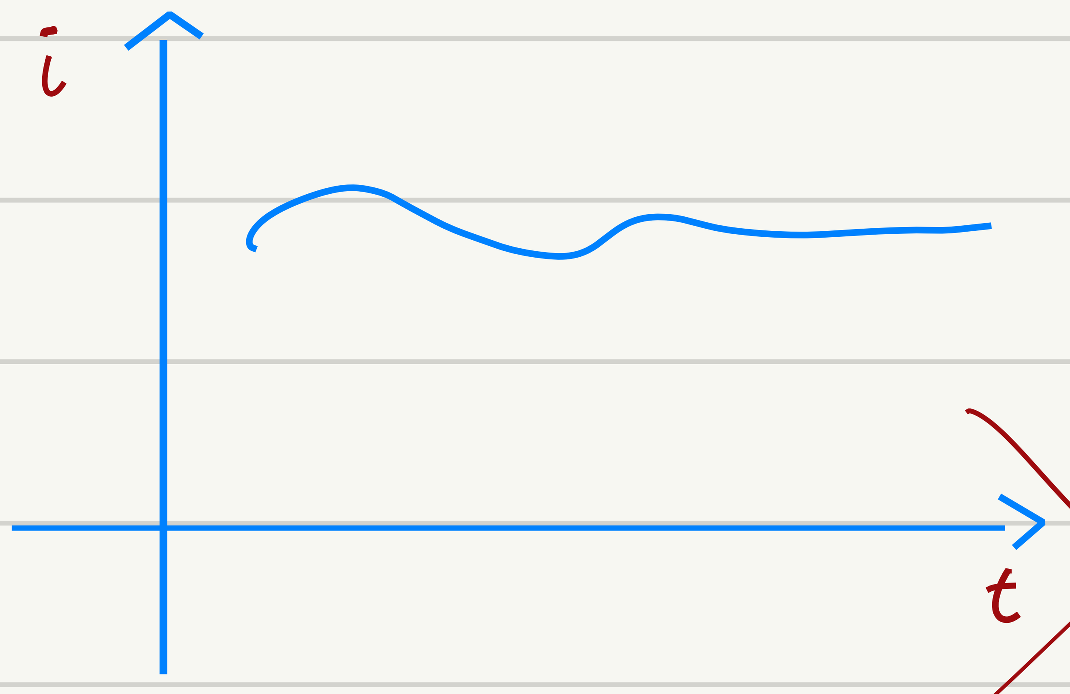
$$\sum_{i=1}^{n} V_i i_i = 0$$

Unit-1 (Introduction) DC circuit

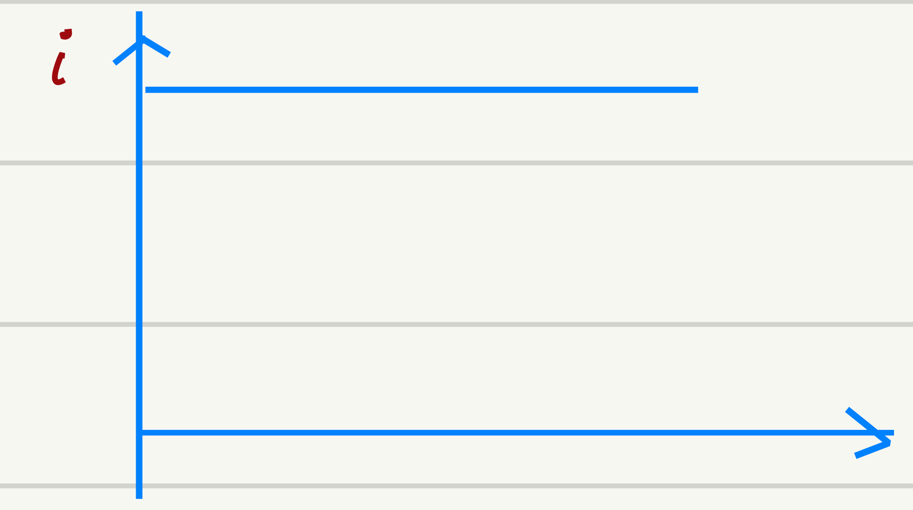
Unit-2

Single Phase AC circuit.

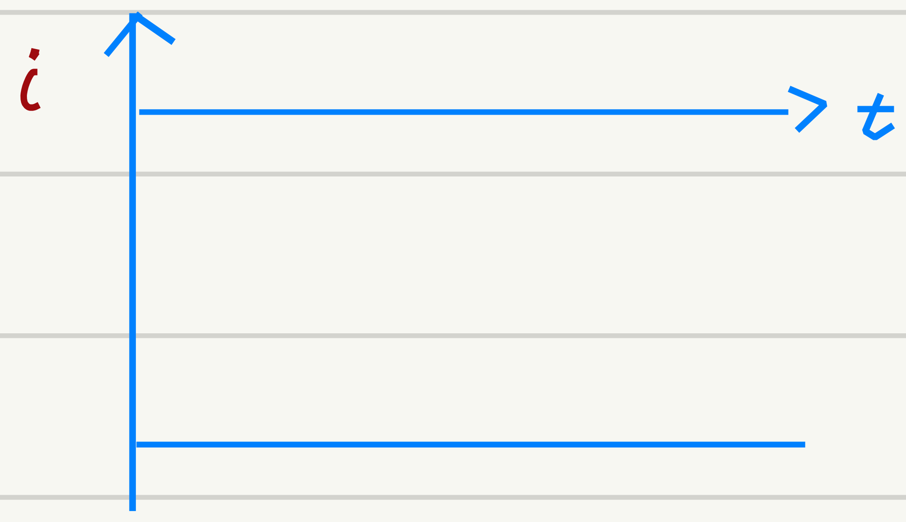
Unit-2 (Single Phase AC circuit)

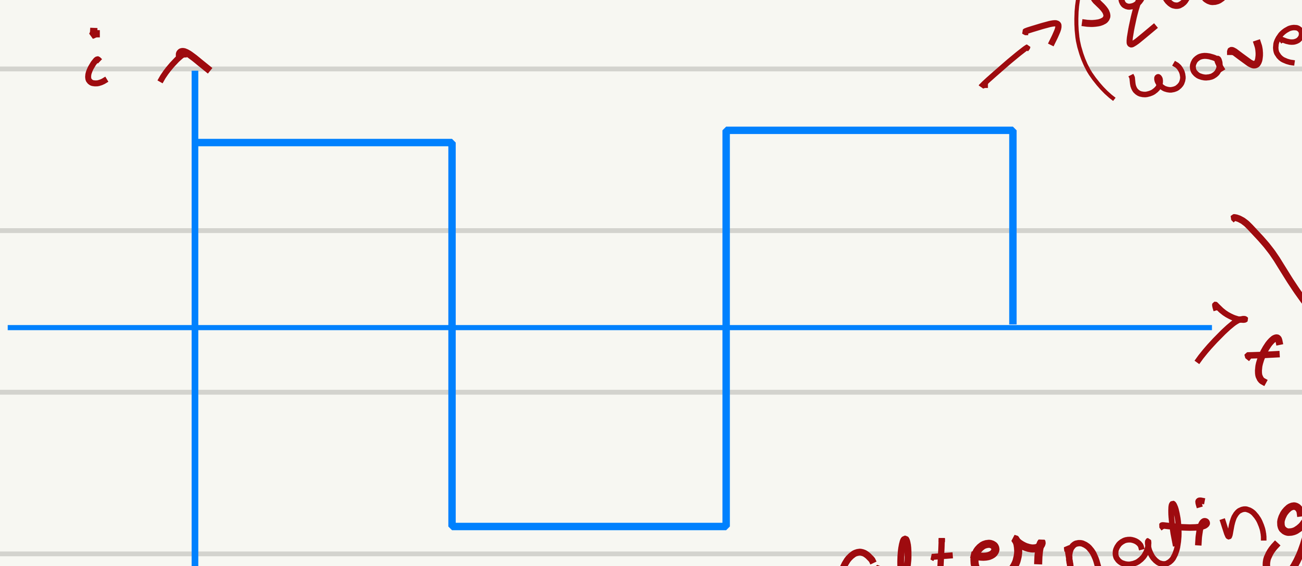


unidirectional current



direct current when i is const. magnitude

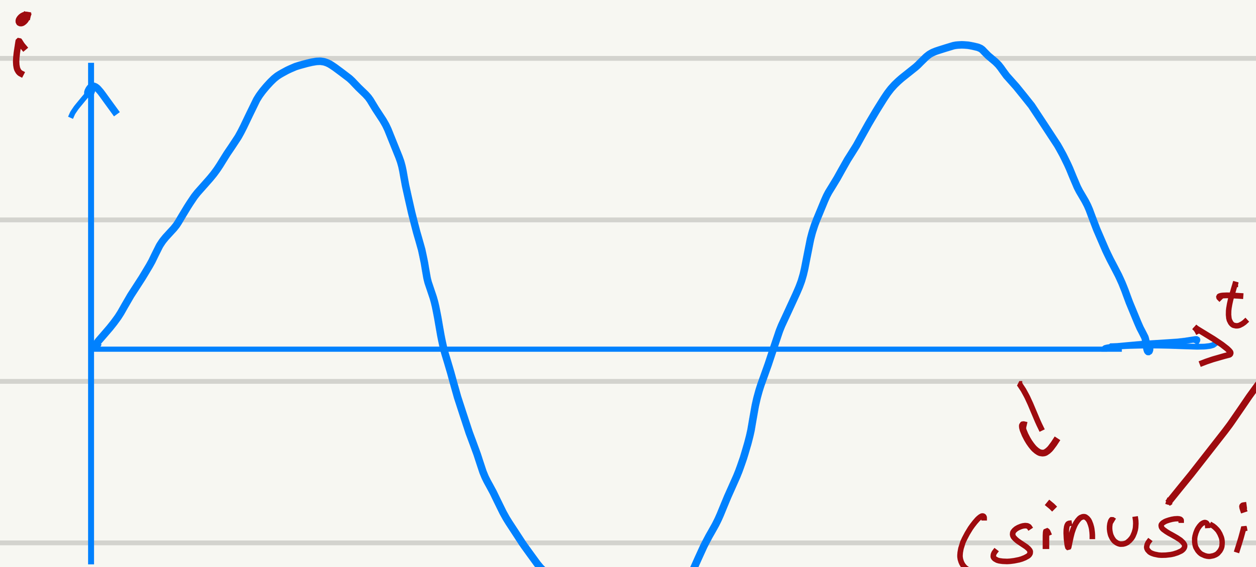




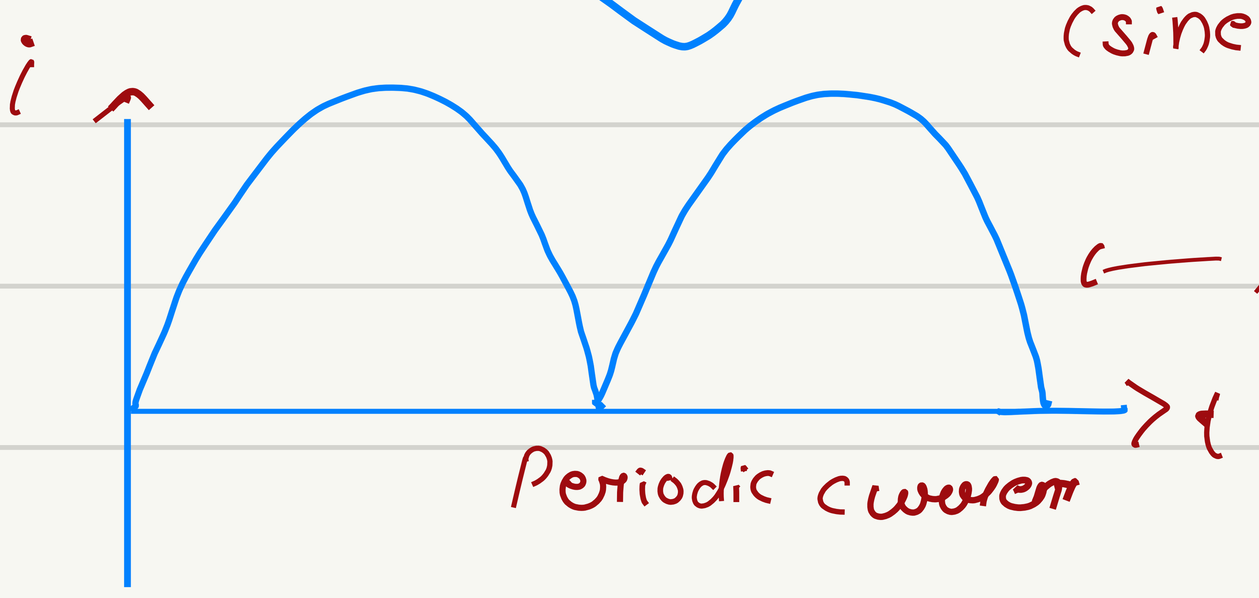
(square wave)

Periodic waveform

alternating current

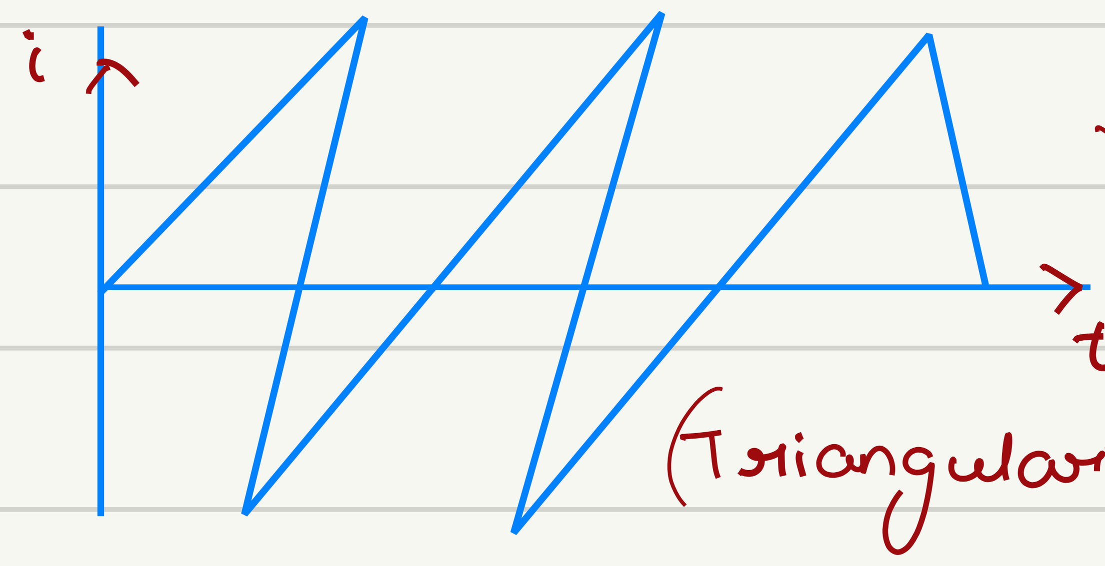


(sinusoidal wave)
(sine wave)



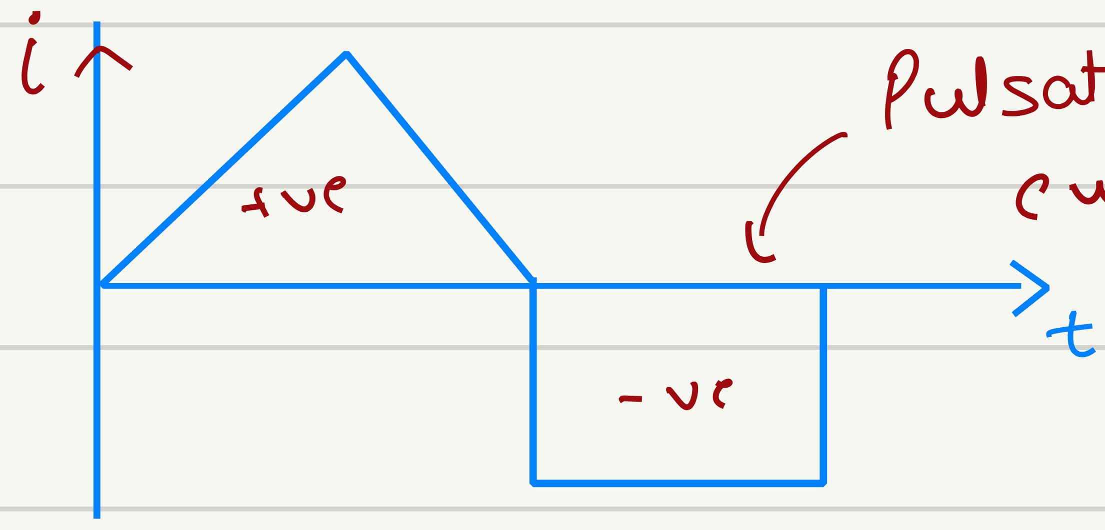
AC to DC

Periodic current



alternating current

(Triangular wave)

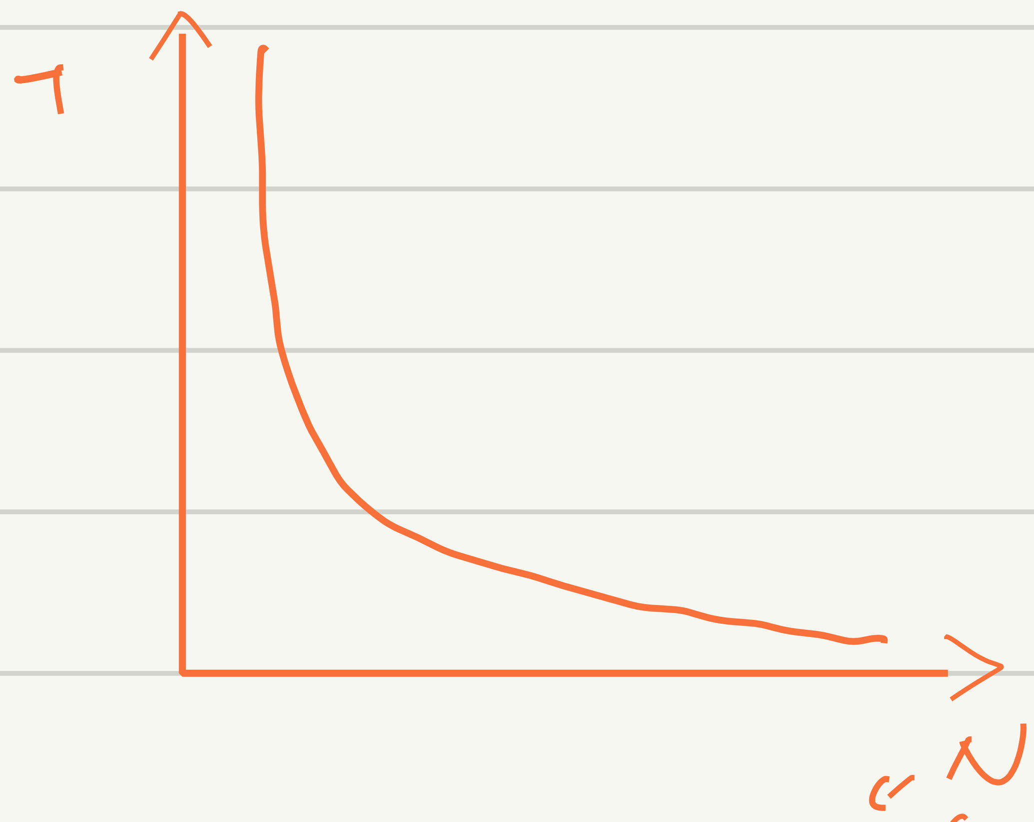


Pulsating current

3000 RPM
1500 RPM
1000 RPM
600 RPM

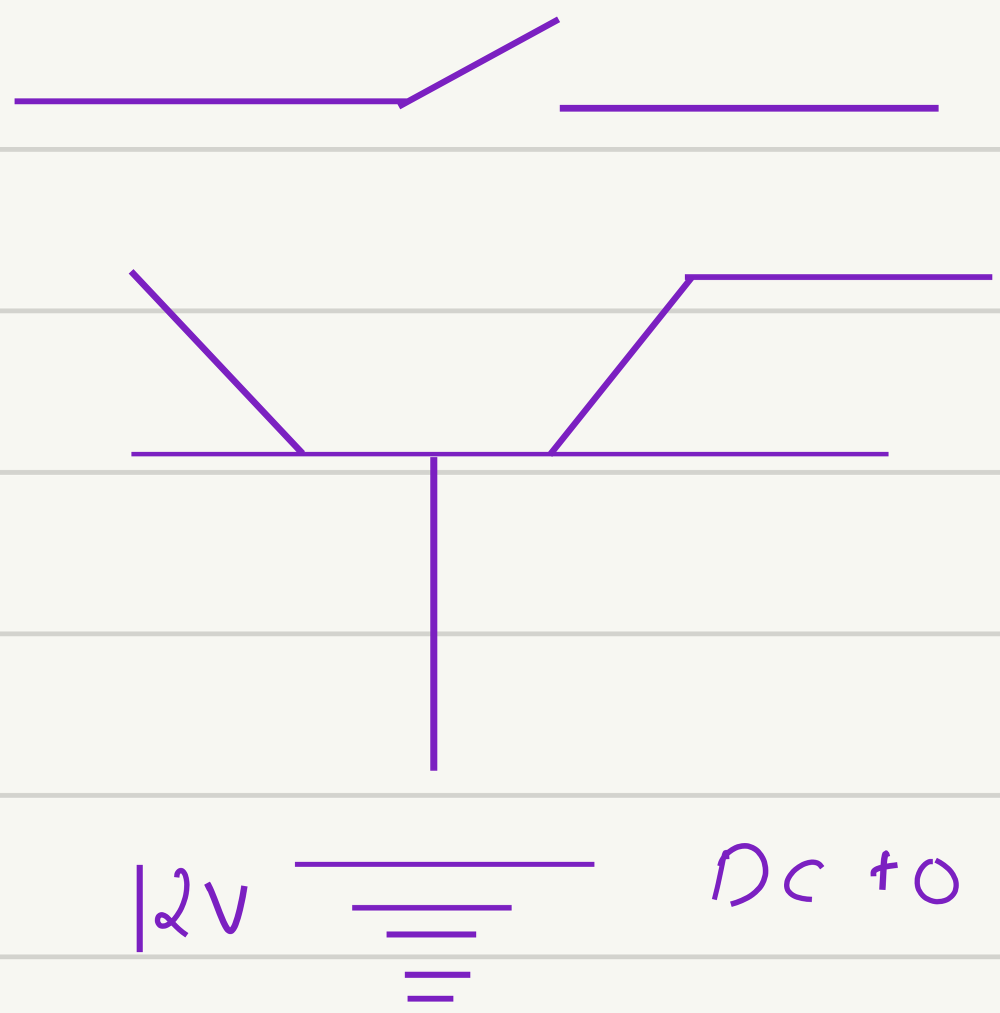
Power Electronics

VSI



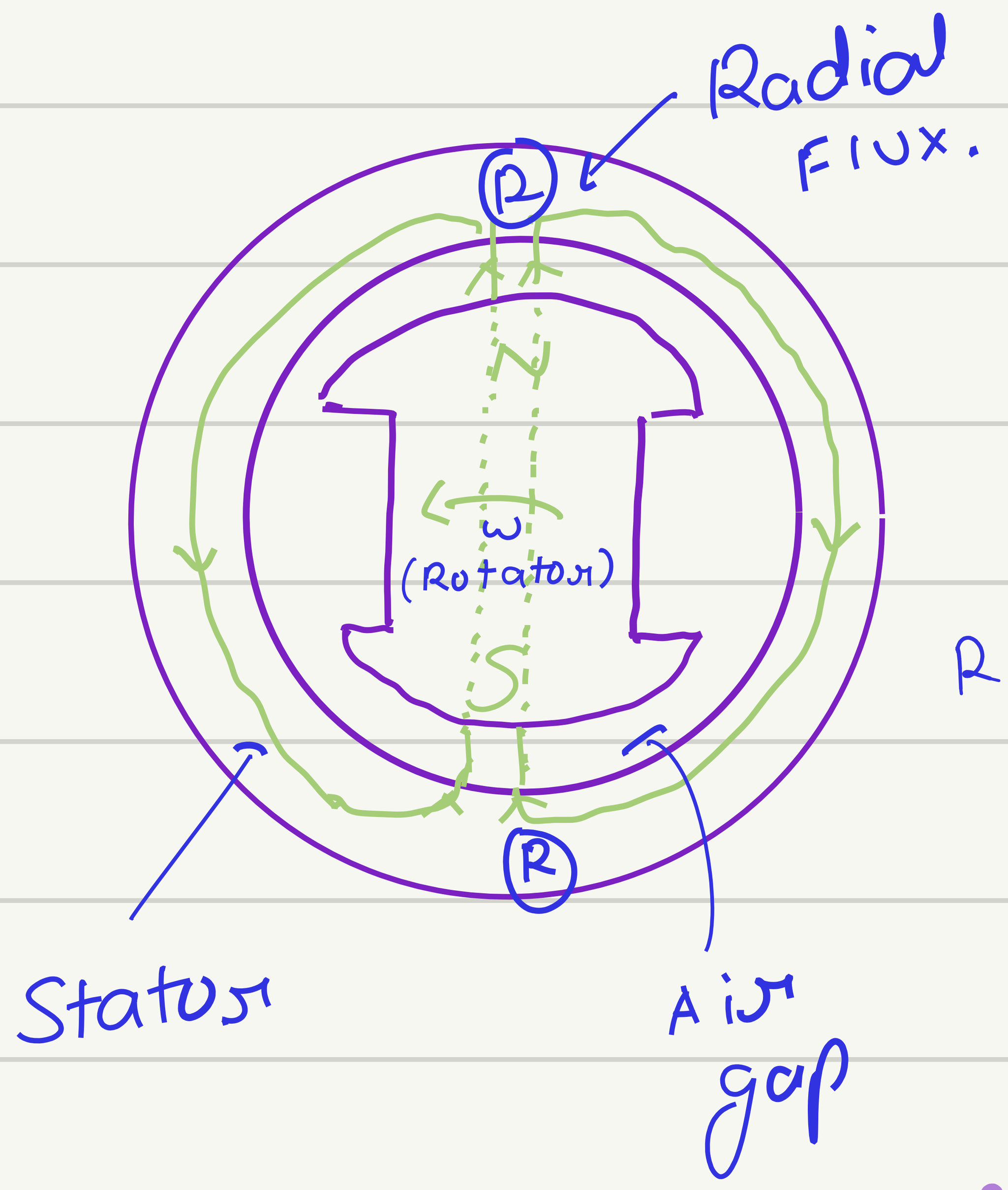
speed (No. of rotation)

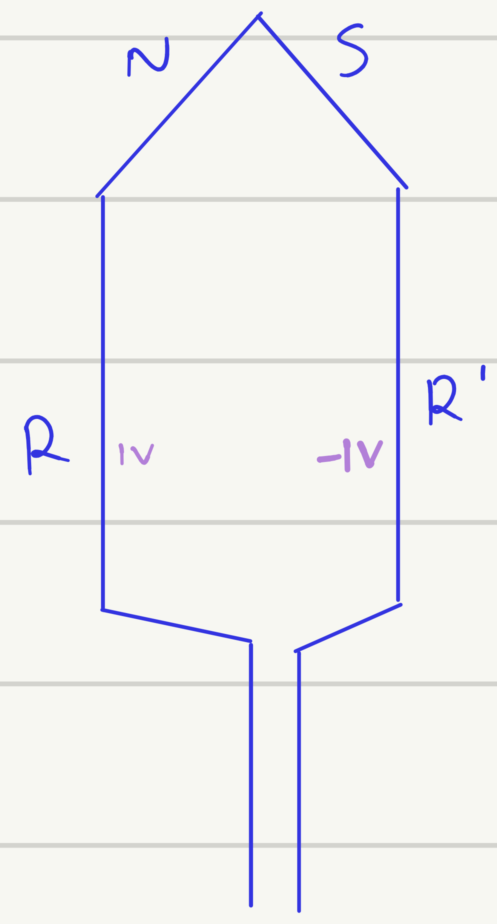
UPS (Uninterupted power supply)
SMPS (Switch mode power supply)



12V

DC to DC





= 2V in one turn × N.

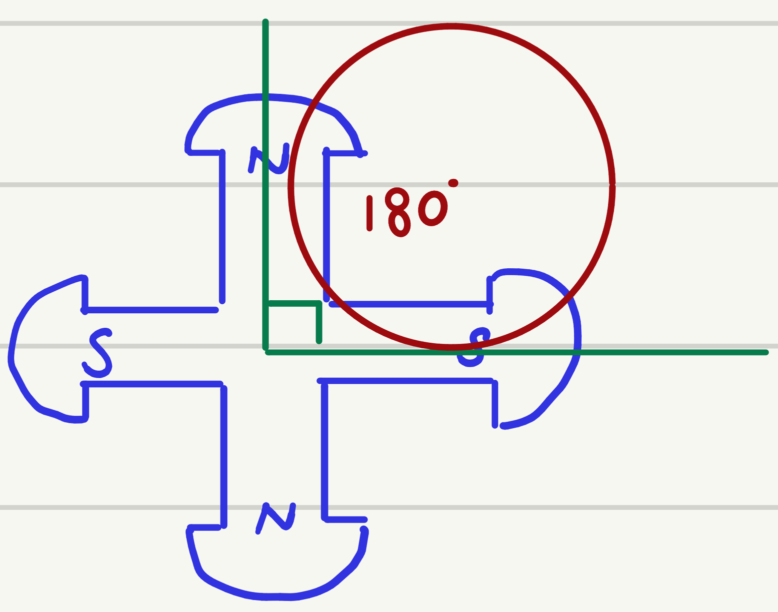
$l \to$ perpendicular
$\phi \to$ radial
$V \to$ tangential

$\phi = \phi_m \cos(\omega t)$
$\omega t \to \theta_e$.

$W_e \to$ Radian/sec
$\theta_e \to$ Radian
$N \to$ no. of turns

$\Psi = N\phi$

$e = \frac{-d\Psi}{dt} = N\omega \sin \omega t$



$\theta_e = \frac{P}{2}\theta_m$

$W_e = \frac{P}{2} W_m$

$\cancel{2\pi} f = \frac{P}{2} \cancel{2\pi} n$

$f \to$ Hz

$n \to$ rps $\to$ revolution

$f = \frac{P}{2} n$

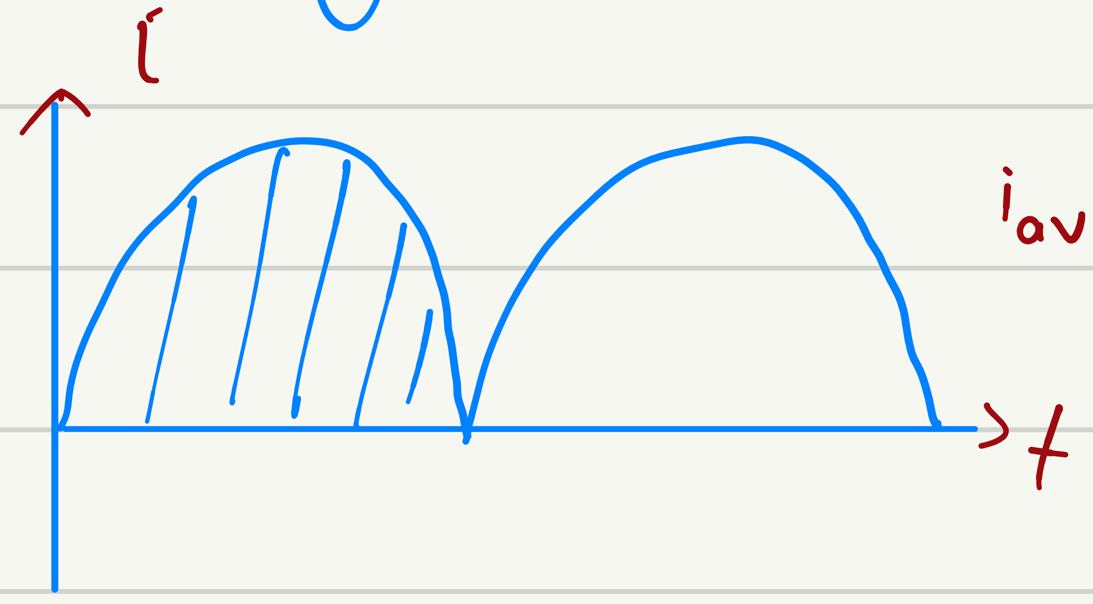
$f = \frac{P}{2}\frac{N}{60}$

50 Hz $\leftarrow$ $\boxed{f = \frac{PN}{120}}$ $\leftarrow$ 3000 RPM

$N \to$ Min

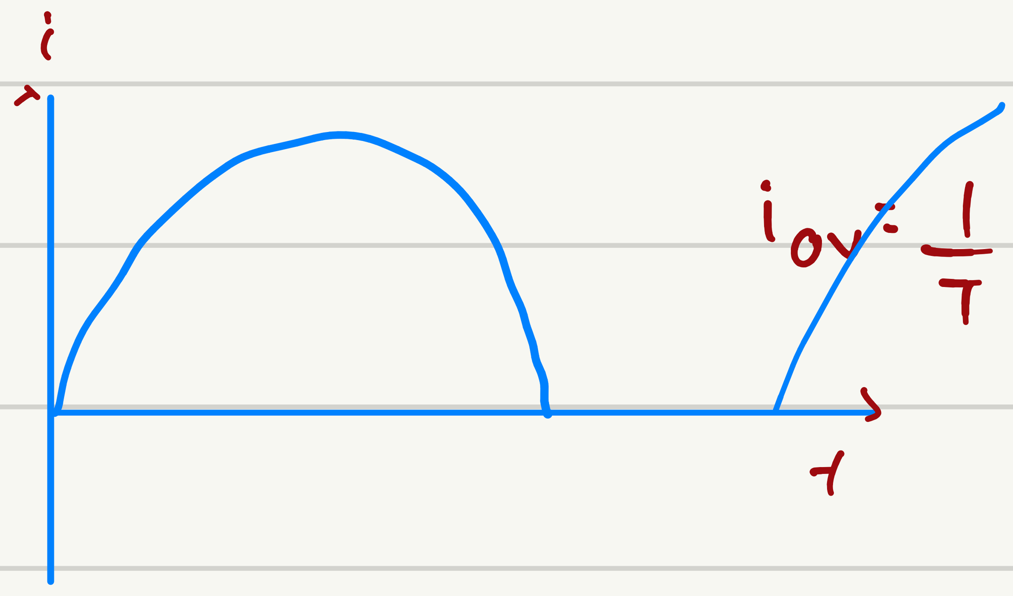
$N = 1500$ RPM for $P = 4$

Average value

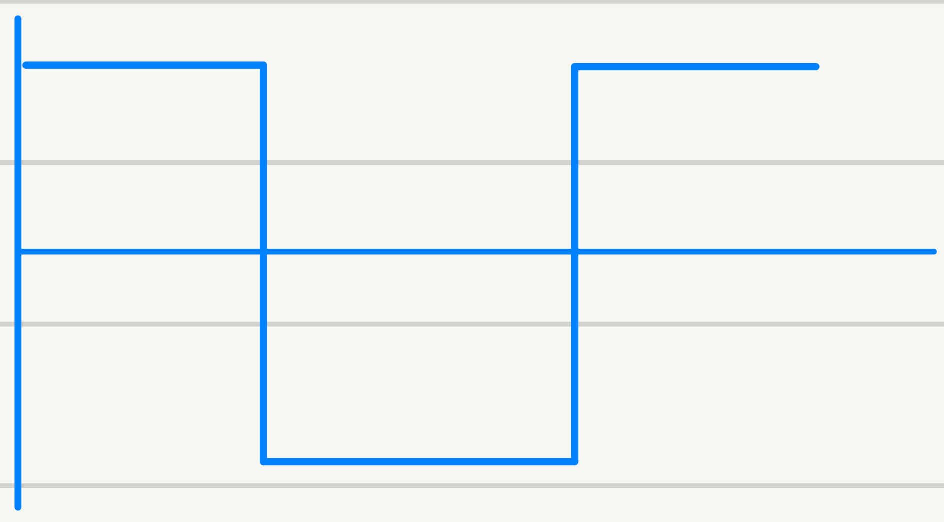


$i_{av} = \frac{1}{T}\int_0^T I_o \sin\omega t \, dt$

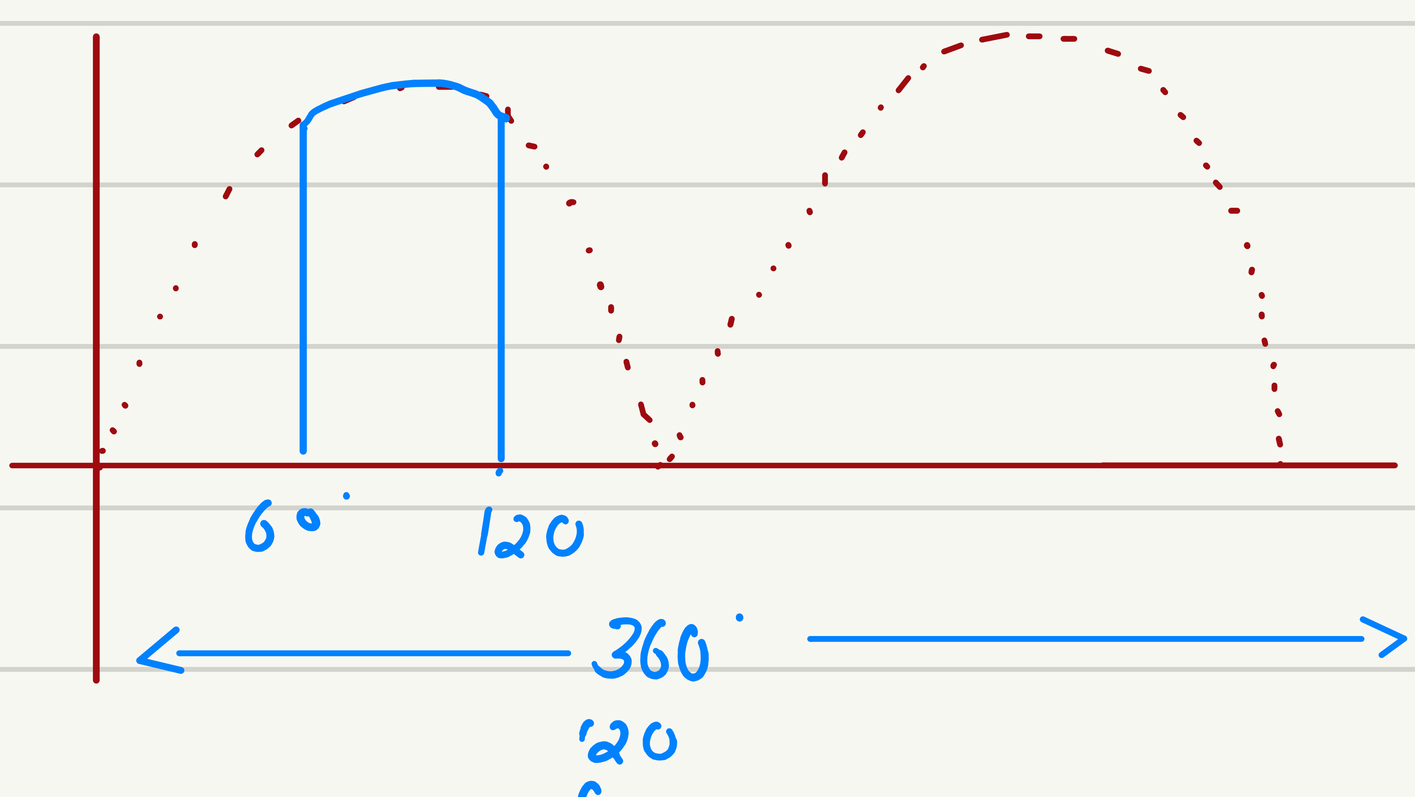
$i_{av} = \frac{1}{T/2}\int_0^{T/2} I_o \sin\omega t \, dt$



$i_{av} = \frac{1}{T}\int_0^{T/2} I_o \sin\omega t \, dt$



$i_{avg} = I_o$



$i_{avg} = \frac{1}{360}\int_{60^\circ}^{120} I_o \sin\omega t \, d(\omega t)$